university of groningen / faculty of arts

The geography of news values in Dutch digital journalism

Mapping newsworthiness in NU.nl coverage

Maartje Pfauder

Master's thesis
Digital Humanities
University of Groningen

https://github.com/MaartjePfauder/Thesis_Digital_Humanities
13-08-2026

Abstract

This thesis examines how news values are geographically distributed in Dutch digital journalism. Using NU.nl articles from 2025, it combines computational text analysis, LLM-assisted classification, and GIS mapping to study how selected forms of newsworthiness become attached to Dutch places. From a corpus of 55,772 articles, 12,225 location-linked articles were classified for six news values: entertainment, bad news, magnitude, good news, celebrity, and power elite. The final GIS dataset contains 19,774 article-location rows and 880 unique geocoded location points.

The results show that NU.nl's location-linked coverage is spatially uneven and concentrated in major urban and institutionally visible places, especially Amsterdam, Rotterdam, The Hague, Utrecht, Groningen, and Eindhoven. Bad news is the dominant form of spatial visibility, but secondary patterns reveal entertainment-oriented places, high-impact magnitude locations, selective good-news visibility, and institutional power-elite visibility. Comparing raw and calibrated scores shows that calibration substantially changes the mapped distribution, particularly by reducing magnitude and increasing bad-news and entertainment presence. The thesis shows how LLM-assisted annotation and GIS can be combined to analyze not only where news refers to places, but how places become newsworthy.

Contents

Preface

This thesis was written as part of the Master's program in Digital Humanities at the University of Groningen. The project grew out of an interest in digital journalism, computational text analysis, and the ways in which media representations shape public understandings of place. I would like to thank my supervisor Scott Eldridge, and everyone else who provided feedback during the development of the thesis.

1 Introduction

Digital news does not represent places evenly. Some cities, towns, and regions appear frequently in national coverage, while others appear only occasionally or mainly in relation to particular types of events. Previous research has shown that news attention is geographically patterned rather than randomly distributed (Özgün, 2023; Vogler et al., 2023). This unevenness matters because journalism does not only report what happens, but also contributes to how places become publicly visible, familiar, problematic, powerful, or culturally significant (Gutsche & Hess, 2019; Usher, 2020). In a digital news environment where articles are continuously published, updated, shared, and circulated, the geography of news is therefore also a geography of mediated attention.

This thesis studies that geography through the concept of news values. News values are recurring criteria of newsworthiness that help explain why some events become news and how published articles construct those events as worthy of attention (Bednarek & Caple, 2017; Harcup & O'Neill, 2017). The focus is not only on where NU.nl articles mention Dutch places, but on the kinds of newsworthiness attached to those places. A city that repeatedly appears in articles about harm, danger, or crime receives a different form of mediated visibility from a place that appears mainly in relation to sport, culture, institutional authority, large-scale consequences, or positive developments.

Existing research provides the separate components needed to study this problem, but has rarely brought them together. News value research has examined how newsworthiness is constructed in published texts and how selected news values can be classified computationally (Bednarek & Caple, 2017; Burggraaff & Trilling, 2020; Cheema et al., 2025; Harcup & O'Neill, 2017). Media-geographical and spatial-bias research has shown that news visibility is geographically uneven (Gutsche & Hess, 2019; Özgün, 2023; Vogler et al., 2023), while digital-humanities research has demonstrated how place references can be extracted, geocoded, and analyzed spatially (Kriesch & Losacker, 2025; Peris et al., 2020; Won et al., 2018). What remains comparatively underdeveloped is the connection between these strands: the spatial distribution of news values themselves, particularly in contemporary Dutch digital journalism. A frequency map can show which places receive attention, but it cannot show through which forms of newsworthiness that attention is produced. This thesis addresses that gap by linking article-level news value classification to extracted Dutch locations and GIS-based spatial analysis.

The main research question is: How are news values geographically distributed in Dutch digital journalism? The question is examined as a case study of NU.nl using the complete set of articles retrieved for 2025 through the Nexis collection procedure used in this project. Operationally, the analysis focuses on six selected news values: entertainment, bad news, magnitude, good news, celebrity, and power elite. The study therefore does not claim to represent every dimension of newsworthiness or every Dutch news outlet, but uses NU.nl to investigate how these selected forms of newsworthiness are geographically patterned in a major Dutch digital news corpus.

The thesis is structured as follows. Chapter 2 reviews literature on news values, digital journalism, media geography, spatial bias, and digital humanities approaches to geolocated news analysis. Chapter 3 describes the methodology, including the corpus, location extraction and geocoding, LLM-assisted news value classification, validation, and GIS preparation. Chapter 4 presents and discusses the spatial results. Chapter 5 concludes by answering the research question, reflecting on the contribution of the thesis, and outlining limitations and future research.

2 Background

This chapter reviews the literature that frames the thesis. It brings together news value theory, digital journalism studies, media geography, research on spatial bias, and digital humanities approaches to computational and spatial news analysis. The chapter first explains why journalism should be understood as a selective practice and why news values can be studied in published texts. It then discusses computational and LLM-assisted classification as a way of connecting theory to method, before turning to the Dutch digital news context, media geography, spatial bias, and the research gap addressed by this thesis.

2.1 The selection of news

Journalism depends on selection. News organizations cannot report all events, statements, issues, and developments, and therefore select some items and ignore others. This selectivity is a basic condition of journalistic work and is shaped by professional judgment, newsroom routines, organizational constraints, available sources, deadlines, audience expectations, and technological conditions (Harcup & O'Neill, 2017; O'Neill & Harcup, 2020). This challenges a simple view of news as a reflection of events. News is connected to events, but it is also the result of recognition, ordering, framing, and presentation. Journalists and editors decide which stories deserve attention, which sources should be used, which details should be foregrounded, and how a story should be placed in relation to other news. These decisions are often made under pressure, as time, access, competition, editorial policy, platform routines, and source availability influence what becomes visible in news output (Harcup & O'Neill, 2017).

News selection is therefore also a matter of exclusion. Some events become publicly visible, while others remain outside the news agenda. Harcup and O'Neill (2017) argue that news values are worth studying because they help explain what is included, what is excluded, and why. At the same time, analyzing news values in published articles has important limits, because news output does not reveal the full editorial process behind selection and publication (Harcup & O'Neill, 2017; O'Neill

& Harcup, 2020). A content analysis of news articles cannot reconstruct editorial negotiations, production routines, or commercial considerations. It can, however, identify recurring forms of newsworthiness as they become visible in published output (Harcup & O'Neill, 2017).

Selection also has spatial consequences. When a place is mentioned in news, it becomes part of a mediated geography. Journalism contributes to the construction of place by repeatedly representing some locations while leaving out others (Gutsche & Hess, 2019; Usher, 2020). Yet visibility alone is too broad as an analytical category. A place can be visible through different types of stories, such as government, crime, court cases, sport, culture, migration, accidents, weather, infrastructure, protest, or local controversy. These forms of visibility carry different meanings. Location frequency can show where news refers to places, but it cannot fully explain the character of that visibility. News value analysis adds an analytical layer by showing through which forms of newsworthiness places become visible.

2.2 News value theory

News value theory starts from the question of what makes something newsworthy. Galtung and Ruge (1965) provide a foundational point of departure because their study of foreign news proposed a systematic set of factors affecting whether events become news, including threshold, unexpectedness, continuity, reference to elite persons, and negativity. Although Galtung and Ruge's framework has been repeatedly revised and qualified in later news value research, the underlying problem remains central to journalism studies: events do not enter the news automatically, but become news through processes of selection, presentation, and interpretation (Harcup & O'Neill, 2017; O'Neill & Harcup, 2020).

Harcup and O'Neill (2017) revisit this tradition and update it for contemporary journalism. Their article builds on their earlier revision of Galtung and Ruge and asks whether older taxonomies remain adequate in a media environment shaped by online news, social media, and audience circulation. Their updated taxonomy contains fifteen news values: exclusivity, bad news, conflict, surprise, audio-visuals, shareability, entertainment, drama, follow-up, the power elite, relevance, magnitude, celebrity, good news, and the news organization's agenda (Harcup & O'Neill, 2017). The taxonomy is useful because it includes both established criteria, such as bad news, conflict, magnitude, relevance, celebrity, and the power elite, and values more closely associated with contemporary digital conditions, such as shareability and audio-visuals.

At the same time, this taxonomy should not be treated as a complete explanation of journalism. O'Neill and Harcup (2020) stress that news values differ by medium, format, publication identity, and context, and that they cannot fully account for all professional, organizational, social, ideological, or subjective factors involved in news production. The taxonomy is therefore best understood as a conceptual starting point rather than as a universal coding scheme that can be applied without adaptation. Mast and Temmerman (2021) make a similar argument about the continued relevance of news value research in the digital age. They argue that news values remain useful, but that their study requires conceptual and methodological care. Their distinction between material, cognitive, social, and discursive dimensions of news values is especially important, because it shows that newsworthiness can be studied from different points in the production, circulation, and reception process.

This conceptual distinction also matters for spatial research. Location mentions show where articles refer to, but they do not by themselves explain how those places are made newsworthy. News value theory adds a vocabulary for asking whether places become visible through negativity, scale, elite actors, cultural appeal, celebrity, or positive developments. In this sense, news values can shape not only which stories appear, but also the geographical pattern through which places become publicly visible.

2.3 News values in published texts

In order to apply news value theory to news output, it is necessary to clarify how news values can be studied in published texts. Because content analysis usually does not have access to internal editorial decision-making, it often approaches news values through the analysis of published articles. The central assumption is therefore that newsworthiness is not only part of editorial selection, but is also constructed and made visible in the way news stories are written, organized, and presented. Harcup and O'Neill (2017) are useful in this respect because their taxonomy is based on the analysis of published news output. Their approach does not claim to reconstruct every editorial decision behind publication. Instead, it shows that recurring criteria of newsworthiness can be identified in news texts.

Bednarek and Caple (2017) provide a more specific theoretical basis for understanding how those values are constructed textually. Their work on the discourse of news values examines how newsworthiness is constructed through language, images, and textual organization. Compared with Harcup and O'Neill, who provide a taxonomy of news values, Bednarek and Caple are more concerned with how those values are constructed in discourse. Their concept of "discursive news values" is useful because it shifts attention from news values as only professional judgments to news values as textual and semiotic constructions. The distinction between cognitive and discursive news values is therefore central. Cognitive news values refer to professional judgments about what is newsworthy, while discursive news values refer to how newsworthiness is constructed in language, images, and textual organization (Bednarek & Caple, 2017; Mast & Temmerman, 2021).

This textual perspective also explains why computational classification is possible. If news values are constructed in discourse, they may leave textual traces that can be identified and classified (Bednarek & Caple, 2017; di Buono et al., 2017). Some news values may leave relatively clear textual traces. Bad news may be constructed through references to death, injury, damage, crisis, loss, or danger; conflict through disagreement, protest, legal dispute, political controversy, or confrontation; and good news through rescue, recovery, success, celebration, or improvement. However, these traces should be treated as indicators rather than definitions. Newsworthiness is produced through context, emphasis, article structure, and interpretation, not through isolated words alone (Harcup & O'Neill, 2017). An article can construct conflict without explicitly using the word conflict, a story can describe a negative event in restrained journalistic language, and a single article can contain more than one news value.

Existing computational studies support this cautious approach. di Buono et al. (2017) examine whether news values can be predicted from headline text and emotion features. Their study shows that computational prediction of news values is possible, but that performance differs across categories. This is relevant because not all news values are equally visible in article text. Values such as bad news, good news, and conflict may be easier to identify from textual features than values such as exclusivity, follow-up, or the news organization's agenda. Their work therefore supports the idea that a news value taxonomy must be adapted to what can be classified with sufficient reliability.

Burggraaff and Trilling (2020) provide a Dutch example of news value operationalization. Their automated content analysis compares online and print news in the Netherlands and operationalizes several news values at scale. Their approach is not based on one universal classifier for all news values. Instead, they use different computational procedures for different values, including named entity recognition, dictionaries, supervised classification, similarity measures, and other automated indicators. This is important because it shows that automated news value analysis can be applied to Dutch news output, including NU.nl, but that operationalization should be tailored to each value rather than treated as a single technical task.

Together, the sources mentioned in this section enable a cautious approach to computational news value analysis. Harcup and O'Neill (2017) provide a taxonomy of newsworthiness in published output, Bednarek and Caple (2017) explain why newsworthiness can be studied as a discursive construction, Mast and Temmerman (2021) clarify that news values in digital journalism require conceptual and methodological care, di Buono et al. (2017) show that computational prediction of news values is possible but uneven across categories, and Burggraaff and Trilling (2020) show that large-scale automated analysis of Dutch online news values is feasible.

2.4 Computational and LLM-assisted classification of news values

The preceding section established that news values can be studied as discursive traces in published news texts. This raises a methodological question: how can such traces be classified across a large corpus of digital news articles? Earlier computational work on news values has shown that computer-based methods can support the analysis of newsworthiness in large datasets. Potts et al.

(2015), for example, demonstrate how corpus linguistic methods can be used to investigate the construction of newsworthiness in a large corpus of news reporting. Piotrkowicz et al. (2017) similarly show that news values can be extracted automatically from headline text and evaluated against manual annotations. These studies support the general premise that news values can be operationalized computationally, while also showing that news value classification remains an interpretive task requiring careful operational definitions and validation.

Large language models offer a relevant extension of this computational tradition. Unlike dictionary-based approaches, which mainly identify predefined words or phrases (Burggraaff & Trilling, 2020), LLM-assisted annotation can account for broader semantic and contextual cues when assigning labels (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024). This is particularly relevant for news value analysis, where recent work has begun to apply LLMs to the automated extraction of news values (Cheema et al., 2025). This matters because newsworthiness is not always expressed through isolated keywords. An article may construct bad news without using the word "bad", or may signal magnitude through scale, consequence, or institutional impact rather than through one fixed expression.

Recent computational social science research supports the use of LLMs for text annotation tasks. Gilardi et al. (2023) find that ChatGPT performs strongly on several annotation tasks, including relevance, stance, topic, and frame detection. Ziems et al. (2024) argue that LLMs can augment the computational social science research pipeline by serving as zero-shot annotators in combination with human oversight. At the same time, they caution that LLMs do not remove the need for validation, especially for taxonomic classification tasks. Törnberg (2024) similarly argues that LLM annotation should follow explicit standards regarding model choice, prompt design, structured outputs, prompt stability, validation, and ethical or legal considerations. LLMs should therefore not be treated as objective measuring instruments, but as annotation tools whose output must be tested against manually coded data.

Cheema et al. (2025) are particularly relevant because they apply LLMs directly to news value analysis. They compare open-source models, including Qwen2, LLaMA3, and Yi-1.5, using structured prompts for news value detection. Their findings suggest that LLM-assisted news value classification is feasible, but that performance depends on model choice, prompting, and validation. They also show that news value annotation remains subjective, as manual annotation produced only moderate inter-annotator agreement. This makes LLM-assisted classification a useful but limited extension of earlier computational work: it can apply a theoretically defined codebook at scale, but it requires validation and careful interpretation.

2.5 The Dutch digital news context

Digital journalism matters for news value research because it shapes the conditions under which news values operate (Mast & Temmerman, 2021). Older news values such as conflict, bad news, relevance, magnitude, and the power elite remain important, but they function within a media environment shaped by continuous publication, online circulation, audience metrics, social media sharing, and platform-specific formats (Trilling et al., 2017). Newsworthiness should therefore be understood in relation to both editorial judgment and the digital infrastructures through which news is distributed.

This need to understand news values in relation to digital conditions is also emphasized by Mast and Temmerman (2021). They argue that news value research must account for changes in the production, circulation, and use of news in digital media environments. Compared with Harcup and O'Neill, who respond to these changes through a revised taxonomy, Mast and Temmerman are more concerned with how news values should be studied conceptually and methodologically in the digital age. Together, these sources show that digital journalism does not require abandoning news value theory, but it does require attention to the changing conditions in which values are applied, circulated, and measured.

The Dutch context matters because NU.nl operates within a specific national media ecology. Vermeer and Groot Kormelink (2024) describe the Dutch media landscape as diverse and characterized by relatively high audience trust, while also noting market concentration and adaptation to digital platforms. Their overview shows that Dutch journalism cannot be understood through one type of outlet alone. Public service media, commercial publishers, regional outlets, digital platforms, and social media all form part of the wider Dutch news environment.

Swart et al. (2017) add an audience-centered perspective to this context. Their study of cross-media news use in the Netherlands shows that people combine different media in everyday news repertoires rather than relying on one stable platform. NU.nl should therefore not be treated as isolated from the wider Dutch news environment. It is a digital platform, but it operates within a cross-media ecology in which audiences encounter news through websites, apps, newspapers, television, radio, social media, and interpersonal circulation.

Within this ecology, NU.nl is a suitable case for studying Dutch digital journalism. It is a national, generalist, continuously updated digital news platform, and its coverage is shaped by platform routines, editorial categories, speed, audience orientation, and national relevance (Burggraaff & Trilling, 2020). NU.nl is not treated as representative of Dutch journalism as a whole, but as a major site through which national digital news visibility is produced. A corpus of NU.nl articles is therefore appropriate for studying how news values operate within a specific digital news environment.

Burggraaff and Trilling (2020) provide an empirical basis for situating NU.nl within Dutch online news research. Their comparison of online and print news includes NU.nl and shows that online news can differ from print news in the prominence of particular values, including follow-up, negativity, positivity, references to persons, and the power elite. Trilling et al. (2017) further connect news values to digital circulation by examining how characteristics of Dutch online news articles, including articles from NU.nl, relate to social media sharing. Their study develops the concept of "shareworthiness" and examines how article characteristics predict sharing on social network sites. Their work shows that traditional ideas of newsworthiness remain relevant in digital environments, but that news values should also be considered in relation to how articles travel online.

Digital publication may also intensify place visibility, as online platforms can publish updates throughout the day. Burggraaff and Trilling (2020) find that online news items are more likely to be follow-up items than print items. Follow-up reporting may repeatedly return to places already connected to newsworthy events. A location associated with one major incident, political decision, legal case, or public controversy may reappear through updates, reactions, consequences, or later developments. Digital journalism can therefore contribute to cumulative place visibility.

2.6 Media geography

Media do not only report on places, but also help organize how places are represented, connected, and understood. The broader framework for this argument comes from communication geography. Adams and Jansson (2012)'s work provides a general bridge between media analysis and spatial analysis. They argue that geography and communication studies should be brought together because communication is always spatially organized. Media connect places, represent places, circulate across places, and shape how places are imagined. This shows that the geographical dimension of news should be treated as embedded in the news itself, rather than as an additional technical layer.

Gutsche and Hess (2019) make this argument more specific to journalism. Their work on geographies of journalism examines how news contributes to the construction of place, space, and territory. Compared with Adams and Jansson's broader communication-geographical perspective, Gutsche and Hess focus more directly on journalistic practice and news representation. They argue that journalism participates in place-making by repeatedly representing locations in particular ways. News media do not simply describe where events happen. They can contribute to collective understandings of places as central, peripheral, familiar, problematic, powerful, or distant.

A further contribution of media geography concerns proximity. In journalism, proximity is often understood as physical distance: events close to the audience are more likely to be considered relevant. Gutsche and Hess (2019) complicate this understanding by linking proximity to power and presence. Their concept of "performative proximity" shows that the significance of a place can also be produced by the presence of powerful actors, institutions, or events. This matters because place visibility in news is not only a question of distance, but also a question of authority, power, and symbolic centrality.

If media geography explains why places in news acquire meaning, it also requires caution about how those meanings are mapped. Usher (2020) focuses on news cartography and journalism's authority over “where”, arguing that place has been under-examined in discussions of journalistic knowledge. This matters because maps of news coverage are not neutral images of space. They depend on choices about article selection, location extraction, classification, geocoding, aggregation, and visualization. Mapping is thus not only a technical output of analysis, but part of the interpretive process through which the geography of news is constructed.

Specht (2018) develops this caution in relation to digital methods. He argues that scholars working with media, communication, and geography should question the data analytics, algorithms, and tools through which space is represented. These tools make large-scale analysis possible, but they also shape the results. A map can suggest precision, while the underlying data may involve ambiguous place names, classification uncertainty, or decisions about spatial aggregation. Together, these perspectives show why media geography is central to the study of news: it explains why spatial distribution matters, while also requiring spatial representations to be interpreted as analytical constructions rather than neutral images of space.

2.7 Spatial bias

Empirical research on the geography of news starts from a simple but important observation: news attention is spatially uneven. News media do not refer to all places with the same frequency or significance. Some places appear regularly because they are politically, economically, culturally, or demographically central, while others appear less often, or mainly when they are connected to specific types of events (Özgün, 2023; Vogler et al., 2023). Spatial bias in news coverage can therefore concern several dimensions at once: how often places are mentioned, which topics they are associated with, which sources are available there, and how close they are to editorial, political, or institutional centers (Özgün, 2023).

Özgün (2023) shows that German news coverage is geographically patterned rather than randomly distributed. Her analysis of this uneven spatial visibility connects regional coverage to factors such as proximity, economic relevance, regional characteristics, and the location of editorial desks or information providers. This shows that geography matters in news content itself and that news coverage should be studied in relation to regional structures, not only in relation to editorial decisions at the level of single articles.

Vogler et al. (2023) approach a related problem from the perspective of geographic diversity. Their study of Swiss regional media examines how different places are represented in news content. Their work is useful because it treats spatial distribution as a measurable dimension of news representation. Together, these studies show that spatial unevenness can be analyzed through patterns of visibility and representation in news content (Özgün, 2023; Vogler et al., 2023). However, these existing studies focus on coverage frequency, geographic diversity, or regional representation. They therefore provide an empirical basis for treating spatial unevenness as a feature of news content, while leaving open the question of how news values are distributed across that uneven geography.

2.8 Digital humanities and spatial news analysis

Having established that spatial unevenness can be analyzed through patterns in news content, it is important to explain how such patterns can be extracted and mapped computationally. Digital and spatial humanities are relevant not only because they provide methods for processing large textual corpora and spatial data, but because such processing involves interpretive decisions about data, tools, and representation (Specht, 2018; Usher, 2020). Transforming articles into classifications, coordinates, and maps involves decisions about what counts as data, how textual meaning is formalized, and how uncertainty is represented. Computational methods should therefore not be treated as neutral instruments, but as ways to analyze spatial patterns at a scale that would be difficult to examine through close reading alone (Peris et al., 2020; Specht, 2018).

A central step in this process is the extraction of place references from text. Peris et al. (2020)'s DIGGER dataset provides a Dutch example. Their project extracts geographical information from 102 million historical news items in Delpher and uses this data to study information diffusion between Dutch cities over more than a century. The relevance of this work lies in demonstrating that Dutch news corpora can be transformed into spatial datasets and used to analyze patterns that would be difficult to identify through close reading alone. However, DIGGER focuses on historical newspapers and long-term information diffusion, rather than contemporary digital journalism or news values. Its relevance is therefore mostly methodological.

Kriesch and Losacker (2025) provide a more recent example of large-scale geolocated news data. Their dataset of German news articles uses named entity recognition for geocoding and includes text embeddings for further semantic analysis. This study is methodologically relevant because it demonstrates how contemporary news data can be prepared for geolocation and semantic analysis. However, it concerns German rather than Dutch news, and its focus is dataset construction rather than the interpretation of news values. It therefore functions as a methodological reference point, not as a direct empirical comparison.

These studies show that news texts can be transformed into geolocated datasets, but they also point to a central methodological issue: spatial analysis is not simply a technical procedure. Place extraction depends on methodological decisions about what counts as a place reference, how ambiguous names are resolved, and how extracted locations are linked to geographic units. Place names may be ambiguous, locations may be confused with organizations or persons, and local or historical spelling variants may complicate recognition. Peris et al. (2020) stress the importance of validation in geographic extraction from news texts. Won et al. (2018) make a similar point in their evaluation of named entity recognition tools for identifying place names in historical corpora. Their work shows that place-name recognition is a central but imperfect step in spatial humanities research.

This concern with validation also applies to computational classification. Digital humanities approaches make large-scale spatial analysis possible, but they also mean that final maps are shaped by classification choices, geocoding decisions, aggregation levels, and visualization design. The methodological value of digital and spatial humanities therefore lies not only in producing maps, but in making the interpretive steps behind those maps visible.

2.9 Research gap

The thesis is situated at the intersection of news value theory, digital journalism studies, media geography, and digital humanities. Each of these fields contributes a necessary part of the project. News value theory provides a framework for analyzing how articles are made newsworthy; digital journalism studies situate NU.nl as a relevant national platform within the Dutch digital news environment; media geography explains why the spatial distribution of news matters; and digital and spatial humanities provide the methodological basis for extracting, classifying, geolocating, and mapping information from large textual corpora.

Taken together, this literature makes the project possible, but it does not yet fully address the geographical distribution of news values in Dutch digital journalism. News value research shows that newsworthiness can be studied in published texts and, to some extent, classified computationally (Bednarek & Caple, 2017; Burggraaff & Trilling, 2020; Harcup & O'Neill, 2017), but does not usually ask to what locations those news values are attached. At the same time, media geography and related studies of spatial bias show that news coverage is geographically uneven and that places can be represented differently in news content (Gutsche & Hess, 2019; Özgün, 2023; Vogler et al., 2023), but often focus on visibility, regional representation, or geographic diversity rather than on the specific news values through which places become visible. Lastly, recent work on LLM-assisted annotation provides a way to operationalize interpretive categories at scale (Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024), but its application to the spatial distribution of news values in Dutch digital journalism remains underdeveloped.

Addressing this gap also requires a methodological bridge between textual analysis and spatial analysis. Digital and spatial humanities provide that bridge by showing how news texts can be transformed into spatial data. Studies on place-name recognition, geocoding, and geolocated news corpora demonstrate that locations can be extracted from large collections of texts and analyzed spatially (Kriesch & Losacker, 2025; Peris et al., 2020; Won et al., 2018). These methods make it possible to study patterns that would be difficult to identify through close reading alone.

The unresolved issue lies in the connection between these strands. Existing research shows that news values matter, that digital news values can be studied computationally, that journalism contributes to place representation, and that news texts can be geolocated. What remains underdeveloped is a spatial analysis of newsworthiness itself. This thesis addresses that gap by treating news values as spatially analyzable features of published digital news. To journalism studies, it contributes an analysis of news values at scale and connects them to space; to media geography, it contributes a way of analyzing place-making through journalistic selection criteria rather than through place mentions alone; and to digital humanities, it contributes an operationalization of a journalism studies concept through computational text analysis, location extraction, and mapping, while keeping the interpretive limits of these methods visible.

To make this connection empirically, the thesis requires a workflow in which news articles, location mentions, article-level news value scores, and spatial units can be linked. The next chapter therefore moves from the theoretical gap identified in the literature to the methodological design of the analysis. It explains how the NU.nl corpus was processed, how Dutch location mentions were extracted and geocoded, how selected news values were classified and validated, and how the resulting article-location data were prepared for GIS-based spatial analysis.

3 Methodology

3.1 Research design

This thesis uses a mixed computational and interpretive research design to examine how selected news values are geographically distributed in Dutch digital journalism. The study combines article extraction, location recognition, coordinate matching, LLM-assisted annotation, benchmark calibration, manual validation, quantitative aggregation, and GIS-based spatial analysis. The aim is not to reconstruct NU.nl's internal editorial decision-making process, but to analyze how selected forms of newsworthiness are detectably present in published article texts and how those forms are attached to Dutch locations mentioned in those texts.

The workflow consisted of three main stages. First, NU.nl articles were converted from Nexis-exported DOCX files into structured article and location tables. Second, the location-linked article subset was classified for six news values using an LLM-assisted annotation procedure. Third, the resulting article-level scores were calibrated, validated, linked to location mentions, and prepared for GIS-based spatial analysis. Both the raw and calibrated scores were retained for the final spatial analysis so that the effect of calibration on geographical patterns could be examined directly.

The method follows a distant reading logic. Rather than analyzing a small number of articles manually, it identifies recurring patterns across the full 2025 NU.nl corpus. This makes it possible to study the relationship between news values and Dutch locations at a scale that would be difficult to examine through close reading alone. At the same time, the method is not treated as a neutral technical process. Each stage involves interpretive decisions: how article text is extracted, which place names are retained, which news values are operationalized, how model outputs are calibrated, and how spatial patterns are aggregated and mapped.

Table 3.1
Computational workflow

Workflow step	Purpose	Analytical output
Article extraction and location matching	Convert NU.nl articles into structured article and location tables.	Article-level corpus table and article-location table.
News value classification and calibration	Assign structured scores for six selected news values and adjust systematic under- or over-detection.	Raw and calibrated article-level news value scores.
Validation and GIS preparation	Compare model scores with manual coding and link article-level scores to geocoded location mentions.	GIS-ready article-location data and raw and calibrated location-level summaries.
Spatial aggregation and mapping	Aggregate raw and calibrated article-location data to points, municipalities, and provinces.	Heatmaps, count maps, raw and calibrated dominance maps, present-share maps, and mean-score checks.

3.2 Corpus and analytical subset

The corpus consists of the complete set of 55,772 Dutch-language NU.nl articles retrieved for the period from 1 January to 31 December 2025. The source material was exported from Nexis as DOCX files and organized by month. During extraction, each article was converted into one structured article row. The extracted fields included article identifiers, titles, publication dates, subject metadata, body text, lead text, text-length indicators, detected locations, and location counts.

The broader article corpus is larger than the spatial analytical subset. Articles without retained

Dutch location mentions remain part of the extracted corpus, but they cannot be used for the GIS analysis because they cannot be linked to a Dutch location. The location extraction workflow initially produced 19,802 location rows. During preprocessing, 28 rows with coordinates outside the European Netherlands bounding box were excluded. The final GIS mention dataset contains 19,774 article-location rows, linked to 12,225 classified articles and 880 unique location-coordinate points.

3.3 Location extraction, cleaning, and geocoding

Dutch locations were identified in article body text using the Dutch spaCy model `nl_core_news_lg`. Candidate entities were retained only when spaCy labelled them as geopolitical entities (GPE) and the entity text matched a location name in an OpenStreetMap-derived lookup file. The OpenStreetMap JSON file was normalized to a table with location names, latitudes, and longitudes. When the lookup contained duplicate location names, the first coordinate was kept. Sixty duplicate OSM location names were identified before this deduplication step. The final lookup contained 2,435 Dutch location names.

The extraction procedure included a preprocessing step to reduce false positives from football club names. Club names that contain city names, such as PSV Eindhoven, Ajax Amsterdam, Feyenoord Rotterdam, FC Groningen, and ADO The Hague, were replaced before location extraction. This was done because such expressions often refer primarily to a football club rather than to the city as a news location.

A stable numeric article identifier was created so that article-level classifications and location mentions could be linked consistently. The structured data were also checked for duplicate article rows and duplicate article-location-coordinate rows. In the final dataset, no duplicate article rows or duplicate article-location rows remained.

The location table was filtered to match the geographical scope of the thesis. Rows with coordinates outside a conservative bounding box for the European Netherlands were excluded. The bounding box used latitude 50.6–53.7 and longitude 3.2–7.3. This removed non-European or otherwise out-of-scope coordinate matches while preserving the focus on Dutch places.

3.4 Units of analysis

The thesis works with two connected units of analysis. The unit of classification is the article. Each classified article receives scores for the selected news values. The primary spatial unit of analysis is the article-location row. A location mention becomes analyzable once it is linked to the article in which it appears and to the news value scores assigned to that article.

Classifying news values at article level is more feasible than classifying every sentence or paragraph around every location mention. It is also consistent with the treatment of news values as properties of how a news story as a whole is constructed as newsworthy and with article-level news value coding in Harcup and O'Neill (2017). The purpose is not to claim that every news value applies independently to every place mentioned in an article. Rather, the spatial analysis measures the co-occurrence of locations with articles containing particular forms of newsworthiness. A secondary or incidental place can therefore inherit a news value that primarily concerns the article's central event or another location. Throughout the thesis, statements that a news value is "attached" to a location should be understood in this representational sense. After classification, each location mentioned in an article inherits the article-level news value scores. This design makes spatial aggregation possible.

3.5 Operationalization of news values

One of the main challenges in this thesis is turning news value theory into a coding scheme that can be applied computationally. This thesis takes the updated taxonomy of Harcup and O'Neill (2017) as its starting point, but focuses on six of the fifteen news values: entertainment, bad news, magnitude, good news, celebrity, and power elite. These values were chosen for both theoretical and methodological reasons. From a theoretical perspective, they capture different ways in which places can become newsworthy, for example through negativity, scale, positive developments, entertainment, well-known individuals, or powerful institutions. (Gutsche & Hess, 2019; Harcup & O'Neill, 2017; Usher, 2020). Methodologically, these values are relatively likely to leave identifiable traces in titles, subject metadata, leads, and article excerpts, which fits the understanding of news values as discursively constructed in published texts (Bednarek & Caple, 2017; Mast & Temmerman, 2021). They are also more feasible for computational and LLM-assisted annotation than values that require editorial context, platform metrics, visual material, or cross-article comparison, such as exclusivity, shareability, audio-visuals, follow-up, or the news organization's agenda (Burggraaff & Trilling, 2020; Cheema et al., 2025; di Buono et al., 2017; Gilardi et al., 2023; Piotrkowicz et al., 2017; Törnberg, 2024; Ziems et al., 2024).

The classification is multi-label rather than multiclass, meaning that One article can contain several news values at the same time. For example, an article about a court case involving a minister and affecting many citizens may contain power elite, bad news, and magnitude simultaneously. The scores therefore do not compete with one another and do not need to add up to 100.

Table 3.2
Working definitions of the six selected news values

News value	Working definition
Entertainment	The article is made newsworthy through amusement, culture, media, leisure, sport spectacle, light human-interest appeal, animals, humor, or entertainment-oriented framing.
Bad news	The article foregrounds harm, loss, death, injury, danger, damage, crime, punishment, crisis, failure, threat, or serious negative consequences.
Magnitude	The article emphasizes scale, large numbers, major financial amounts, broad consequences, national or regional impact, records, exceptional scale, or events affecting many people.
Good news	The article foregrounds success, victory, achievement, recovery, rescue, improvement, celebration, breakthrough, positive return, favorable outcome, or a problem being solved.
Celebrity	The article is made newsworthy through famous or widely recognized individuals, including athletes, royals, artists, presenters, influencers, entertainers, or other public figures.
Power elite	The article foregrounds powerful actors or institutions, such as government, parliament, courts, police, municipalities, ministries, regulators, major companies, universities, hospitals, or other major organizations.

The definitions were implemented in a prompt codebook. The full classification prompt is reproduced in Appendix B. The codebook gave the model definitions, typical indicators, exclusion rules, and scoring instructions. The indicators were not used as a rigid dictionary. Instead, they guided the model in applying the definitions to article excerpts while allowing contextual judgment.

3.6 Scoring system

Each news value was coded using a three-level presence scale: 0, 50, and 100. The purpose of the scale was to produce standardized article-level scores that could be aggregated and mapped.

Table 3.3
Three-level scoring scheme for news value classification

Label	**Numeric score**	**Meaning**
Absent	0	The news value does not meaningfully contribute to the article's newsworthiness.
Secondary	50	The news value is present, but secondary or not the main reason the article is newsworthy.
Central	100	The news value is central to why the article is newsworthy.

The values 0, 50, and 100 are numerical encodings of three ordinal categories—absent, secondary, and central—rather than continuous measurements. A score of 50 does not mean that a news value constitutes 50% of an article, and a score of 100 does not represent complete certainty or twice the empirical quantity represented by 50. The numerical coding was used to make the ordered categories compatible with aggregation and GIS visualization. Calculating mean scores therefore provides a summary index of how observations are distributed across the three ordered categories and relies on the pragmatic assumption of equally spaced coding steps. It should not be interpreted as measurement on a naturally continuous interval scale. For this reason, the analysis also reports present shares, which collapse the scores into the simpler distinction between absent and present and provide an additional check on the spatial patterns.

3.7 LLM-assisted news value classification

The main classification method is LLM-assisted annotation. The model was used as a structured annotation tool that applied the predefined codebook to article excerpts. The open-weight instruction model `Qwen/Qwen2-1.5B-Instruct` was used in Google Colab. This model choice was informed by Cheema et al. (2025), who compared Qwen2, LLaMA3, and Yi-1.5 for automated news value detection and found that Qwen2 achieved the best overall results among the evaluated models. Their study also supports the use of structured prompts and JSON-style outputs for news value annotation, while emphasizing that LLM-based classification should still be validated against human coding (Cheema et al., 2025; Törnberg, 2024; Ziems et al., 2024).

For each article, the model received a compact classification prompt containing the numeric article ID, publication date, title, subject metadata, detected Dutch locations, the text source used, and an article excerpt. The excerpt was built from the lead text when the lead was available and sufficiently informative. If the lead was shorter than 600 characters and body text was available, the lead and body text were combined. If the lead was missing or shorter than 80 characters, the body text was used. The excerpt was then truncated to 750 characters. The use of a standardized excerpt rather than the full article was both a computational and an analytical choice. During the pilot, it became clear that processing the complete location-linked corpus was computationally demanding. Even with the shortened inputs, classifying the full set of 12,225 articles required days. The relatively small open-weight model used in the study also made controlled prompt length desirable. Standardizing the amount of article text reduced variation in input length between short and long articles and made the classification procedure more consistent across the corpus. In addition, the analysis was designed to identify prominent and foregrounded forms of newsworthiness rather than every possible news value mentioned anywhere in an article. Leads were therefore preferred when sufficiently informative, with body text added or substituted when necessary. The consequence is that news values expressed only later in an article may be missed, which is treated as a limitation of the method.

The prompt instructed the model to classify all six values independently and to return a structured

JSON object with one score per news value, a dominant-value field, and a short reason. Scores were restricted to the valid values 0, 50, and 100. Invalid JSON or invalid score values triggered a retry. If parsing still failed after the maximum number of retries, the article received zero scores and was marked with a parse error. The final full classification contained 12,225 classified articles and had a parse success rate of 0.988.

The classification workflow was tested before the full run. A small smoke test checked whether the model loaded correctly, whether the prompt returned parseable output, and whether score validation worked. A pilot classification of 150 articles was then used to inspect whether the operational definitions and the three scoring categories were being applied plausibly across different types of articles. Prompt wording was adjusted iteratively during this development stage. However, the pilot was not designed as a formal prompt-sensitivity experiment: alternative category orderings, example sets, exclusion rules, or prompt formulations were not systematically compared using separate quantitative evaluation sets.

The final classification prompt used four manually coded few-shot examples to demonstrate the intended application of the codebook and the 0/50/100 scoring scheme. These examples were drawn from the manually coded few-shot examples used in the classification workflow and represented different combinations of news values and score levels. The final prompt template is reproduced in Appendix B, while the four manually coded few-shot examples are documented in Appendix C.

3.8 Benchmark prevalence calibration

The workflow includes a benchmark prevalence calibration step. This was added because the raw model output strongly under-identified some values, especially celebrity and entertainment, and over-identified magnitude. Rather than changing the conceptual definitions, the workflow applied a corpus-level adjustment so that the final prevalence of the six selected news values approximated the empirical distribution reported by Harcup and O'Neill for newspaper page-lead stories. For the six values used in this thesis, the benchmark counts were converted into target presence shares and applied independently to each news value.

Table 3.4

Benchmark prevalence shares used for calibration

News value	Benchmark count	Benchmark share
Bad news	442 / 711	62.2%
Entertainment	332 / 711	46.7%
Power elite	216 / 711	30.4%
Magnitude	165 / 711	23.2%
Celebrity	145 / 711	20.4%
Good news	137 / 711	19.3%

Calibration was carried out separately for each of the six news values after the raw Qwen2 scores had been generated. For each news value v and article i, a ranking score was calculated as:

$$R_{iv} = 100S_{iv}^{\text{raw}} + K_{iv}, \tag{1}$$

where S_{iv}^{raw} is the original Qwen2 score (0, 50, or 100), and K_{iv} is the number of predefined calibration keywords for that news value found in the article context. This context included the classification text, title, subject metadata, and the short explanation produced by the model. The Qwen2 score was given much greater weight than the keyword count, meaning that the model's classification remained the main basis for the ranking. Keyword information was used only to distinguish between articles with the same or similar raw scores.

For each news value, the benchmark share reported in Table 3.4 was used to determine the target number of articles in which that value should be present. Articles were then ranked according to the score described above. When articles received the same ranking score, the raw Qwen2 score was used as a second criterion. A deterministic hash of the article ID was used only as a final tie-breaker, ensuring that the same articles would be selected if the procedure was repeated.

The highest-ranked articles were selected until the benchmark target for each news value was reached. Their calibrated scores were then assigned on the basis of the original Qwen2 classification.

Articles with a raw score of 50 or 100 retained that score. If an article originally received a score of 0 but was included through the calibration procedure, it was assigned a score of 50. In this way, calibration could introduce a news value as a secondary feature, but could not create new cases in which that value was considered strongly present. Articles that fell outside the benchmark target received a calibrated score of 0. The original Qwen2 scores were retained separately in the `raw_*` columns, allowing the raw and calibrated results to be compared directly.

This calibration introduces an external assumption about how common each news value should be. The benchmark is based on Harcup and O'Neill (2017)'s analysis of newspaper page-lead stories, whereas the present study examines NU.nl articles published in 2025. The benchmark should therefore not be interpreted as the true distribution of news values on NU.nl. Instead, it was used as a reference point to correct clear imbalances in the raw model output, particularly where certain categories appeared to be strongly under- or over-detected. Importantly, the benchmark was applied independently of the manually coded validation sample. The raw and calibrated scores were subsequently evaluated against the same manual annotations.

3.9 Manual validation and comparison of raw and calibrated scores

The thesis also includes manual validation. A validation sample of 300 articles was manually coded using the same 0/50/100 scoring scheme as the model. All 300 validation articles received complete manual scores for the six selected news values. The validation compared the calibrated model scores with the raw uncalibrated scores.

The validation reports both exact three-level agreement and binary present/absent metrics. Exact agreement evaluates whether the model selected the same score as the manual annotation: 0, 50, or 100. Binary precision, recall, and F1-score convert scores above 0 into present and scores of 0 into absent.

Across the six news values, the calibrated version achieved a mean exact three-level accuracy of 0.764, a mean absolute error of 13.4 points on the 0–100 scale, a mean Cohen's kappa of 0.458, and a mean present/absent F1-score of 0.691. The raw version achieved a mean exact accuracy of

0.757, a mean absolute error of 15.5, a mean Cohen's kappa of 0.347, and a mean present/absent F1-score of 0.517. Calibration therefore slightly improved exact accuracy, reduced average error, and substantially improved present/absent classification.

Table 3.5

Manual validation metrics and present shares for raw and calibrated model output

News value	Exact acc.	MAE	Kappa	Precision	Recall	F1	Manual present	Raw present	Calibrated present
Entertainment	0.650	20.0	0.436	0.777	0.831	0.803	43.3%	12.0%	46.3%
Bad news	0.670	17.8	0.478	0.631	0.965	0.763	38.3%	27.3%	58.7%
Magnitude	0.867	7.0	0.549	0.679	0.792	0.731	16.0%	33.7%	18.7%
Good news	0.853	10.0	0.384	0.447	0.656	0.532	10.7%	9.7%	15.7%
Celebrity	0.833	8.5	0.527	0.815	0.579	0.677	25.3%	2.0%	18.0%
Power elite	0.710	17.0	0.375	0.682	0.606	0.642	33.0%	27.7%	29.3%

Note. Exact accuracy, MAE, Cohen's kappa, precision, recall, and F1 refer to the calibrated model scores. Scores above 0 were treated as present for precision, recall, F1, and the three present-share columns.

The validation shows why calibration could be preferable for the GIS analysis. The raw model strongly under-detected entertainment and celebrity. Entertainment was manually present in 43.3% of validation articles, but the raw model predicted it in only 12.0%. After calibration, the predicted present share was 46.3%. Celebrity was manually present in 25.3% of articles, while the raw model predicted it in only 2.0%; after calibration this increased to 18.0%. Magnitude moved in the opposite direction: the raw model predicted magnitude in 33.7% of validation articles, compared with a manual share of 16.0%, while the calibrated version predicted 18.7%.

Calibration did not improve every category equally. Bad news had very high recall after calibration, 0.965, but lower precision, 0.631. Good news remained the weakest category after calibration, with an F1-score of 0.532. It should therefore be interpreted more cautiously than entertainment, magnitude, celebrity, and power elite.

The article-level disagreement analysis also shows that calibration improved total error for 113 articles, worsened total error for 53 articles, and made no difference for 134 articles. This could be seen as support for using the calibrated scores as the preferred analytical version. However, because calibration itself changes the prevalence of several news values, the raw scores were retained as a comparison condition in the GIS analysis. The final spatial analysis therefore compares raw and calibrated patterns directly.

3.10 GIS-ready datasets and location-level summaries

After classification and calibration, the article-level news value scores were merged with the location-mention table. This created the main GIS-ready article-location dataset. Each row represents one retained Dutch location mentioned in one classified article and contains the location name, coordinates, article metadata, raw and calibrated news value scores, and diagnostic classification information.

The final GIS mention dataset contains 19,774 article-location rows, linked to 12,225 classified articles and 880 unique location points. No rows in the final GIS dataset have missing coordinates, no rows fall outside the European Netherlands bounding box, and no rows are missing classification scores. A small number of article classifications were marked with parse errors. In the final GIS-ready file, this concerns 186 article-location rows, corresponding to 147 unique articles. These rows remained in the downstream GIS dataset because the original workflow encoded repeated parsing failures using fallback zero scores. They are therefore retained as part of the implemented workflow but are treated as technical failures rather than as substantive all-zero classifications; their possible effect on the analysis is discussed in Section 4.6.

For broader interpretation, parallel raw and calibrated location-level summaries were created. For each location-coordinate combination, these summaries include the mention count, mean score, present share, strong share, and sum score for each of the six news values. The mean score describes the average intensity of a news value across all mentions of a location. The present share describes the proportion of mentions in which the news value is present at either 50 or 100. The strong share describes the proportion of mentions in which the news value is central, scored as 100. The raw and calibrated summaries use the same locations, coordinates, mention counts, and aggregation procedure, so differences between them reflect only the use of raw versus calibrated news value scores.

The summaries also identify a dominant news value at location level based on the highest mean score. This field provides a compact description of the news value with the highest average score for each geocoded location and allows raw and calibrated location profiles to be compared directly.

3.11 QGIS setup and spatial data sources

The spatial analysis was carried out in QGIS. The aim of the GIS workflow was to examine how raw and calibrated news value scores are distributed across Dutch locations, municipalities, and provinces, and to determine how the calibration procedure changes the resulting spatial patterns. The GIS analysis therefore distinguishes between three spatial levels: geocoded point locations, municipality-level aggregation, and province-level aggregation.

Two types of GIS-ready data were used as input. The detailed article-location layer contains one row per article-location combination and includes both raw and calibrated news value scores. This layer was used for the overall location mention heatmap and for mention-weighted municipality and province aggregation. It was also used to calculate municipality-level raw and calibrated present shares and mean scores. Location-level summary layers contain one row per unique geocoded location and were used for the mention-count map and location-level dominant-news-value comparisons.

The point data were originally stored with longitude and latitude coordinates. They were imported into QGIS using longitude as the X coordinate, latitude as the Y coordinate, and WGS 84 as the source coordinate reference system. After import, the layers were saved as GeoPackage layers and reprojected to Amersfoort / RD New (EPSG:28992).

Administrative boundary layers for municipalities and provinces were downloaded from the PDOK/Kadaster Bestuurlijke Gebieden service. These boundary layers were used only for spatial aggregation of the news value point data.

3.12 GIS map set and spatial aggregation

The GIS workflow was designed to move from extracted point locations to broader administrative patterns while allowing direct comparison between the raw and calibrated classification results. The final map set combines baseline visibility maps, dominant-news-value maps, and present-share maps.

Table 3.6
GIS map set and analytical purpose

Map set	Main method	Analytical purpose
Overall location mention heatmap	Heatmap of article-location rows	Baseline density of retained Dutch location mentions.
Mention count by location	Graduated point size by mention count	Volume of visibility per unique geocoded place.
Dominant news value by location point	Highest average news value score per location	Exploratory point-level overview and comparison of raw and calibrated place profiles.
Dominant news value by municipality	Mention-weighted mean raw and calibrated scores	Compare full and no-bad-news municipal dominance before and after calibration.
Dominant news value by province	Mention-weighted mean raw and calibrated scores	Compare full and no-bad-news province-level patterns before and after calibration.
Present-share maps by municipality	Mean of binary presence fields for raw and calibrated scores	Compare how frequently each of the six news values is present before and after calibration.

The overall heatmap and mention-count map distinguish the volume of location visibility from the composition of news values attached to those locations. The heatmap shows where Dutch article-location mentions cluster, while the mention-count map shows how often each unique geocoded place appears. Together, these maps provide a baseline view of spatial attention before news values are interpreted.

Dominant-news-value maps were used to show which news value had the highest mean score for each spatial unit. At municipality and province level, the final analysis uses mention-weighted aggregation. Each article-location row contributes one observation, so frequently mentioned places have proportionally more influence on the aggregated mean scores. The same procedure was applied separately to the raw and calibrated score fields, allowing the spatial effect of calibration to be assessed directly.

Because bad news is prominent in the calibrated data and frequently dominates the full maps, additional no-bad-news dominance maps were created for both the raw and calibrated results. These maps do not remove articles containing bad news from the dataset. Instead, bad news is excluded only from the set of candidate values used to determine the dominant category. The maps therefore show which of the remaining five news values has the highest mean score when bad news is not allowed to determine dominance.

The municipality-level dominance analysis was restricted to municipalities with at least ten article-location mentions. This threshold reduces the influence of highly unstable results based on very small numbers of observations. The same threshold was applied to both the raw and calibrated municipal maps so that the two versions contain the same set of municipalities and can be compared directly. Province-level dominance was calculated across all twelve provinces using the same mention-weighted raw and calibrated scores.

Present-share maps were created separately for all six news values using both the raw and calibrated scores. Each score was first converted into a binary presence field: scores of 50 or 100 were coded as present and scores of 0 as absent. The municipal present share is therefore the mean of this binary field and represents the proportion of article-location mentions within a municipality in which a particular news value is present. The same minimum threshold of ten article-location mentions was applied to the raw and calibrated present-share maps.

4 Results and Discussion

This chapter presents the results of the spatial news value analysis. It begins with the overall distribution of Dutch location mentions in the NU.nl corpus, before turning to the news values attached to those locations. The analysis first establishes where location-linked coverage is concentrated and then compares the spatial patterns produced by the raw and calibrated news value scores. This comparison makes it possible to examine not only the final calibrated geography of newsworthiness, but also how calibration changes the relative prominence of different news values across the corpus and across municipalities.

4.1 Corpus overview

The raw and calibrated corpus-level distributions of the six news values are compared in Table 4.1. The comparison shows that calibration changes the prevalence of several categories substantially. These differences are important for the spatial analysis because shifts in corpus-level prevalence can also affect which news value becomes dominant after scores are aggregated geographically.

Table 4.1
Raw and calibrated corpus-level distribution of news values

	Present articles		Present share		Mean score		Strong articles		Strong share	
News value	**Raw**	**Cal.**	**Raw**	**Cal.**	**Raw**	**Cal.**	**Raw**	**Cal.**	**Raw**	**Cal.**
Bad news	3,553	7,600	29.1%	62.2%	29.1	45.6	3,553	3,553	29.1%	29.1%
Entertainment	1,545	5,708	12.6%	46.7%	12.6	29.6	1,533	1,533	12.5%	12.5%
Power elite	3,496	3,714	28.6%	30.4%	16.2	17.1	475	475	3.9%	3.9%
Magnitude	4,469	2,837	36.6%	23.2%	21.1	14.4	687	687	5.6%	5.6%
Celebrity	426	2,493	3.5%	20.4%	2.9	11.3	276	276	2.3%	2.3%
Good news	1,499	2,356	12.3%	19.3%	12.0	15.5	1,444	1,444	11.8%	11.8%

The raw results show a different distribution from the calibrated results. Magnitude has the highest raw present share at 36.6%, followed by bad news at 29.1% and power elite at 28.6%. Entertainment and good news are present in 12.6% and 12.3% of articles respectively, while celebrity is detected in only 3.5%. After calibration, bad news becomes the most prevalent value at 62.2%, followed by entertainment at 46.7%. Celebrity also increases substantially to 20.4%, while good news rises to 19.3%. Power elite changes comparatively little, from 28.6% to 30.4%. Magnitude moves in the opposite direction, decreasing from a raw present share of 36.6% to a calibrated share of 23.2%.

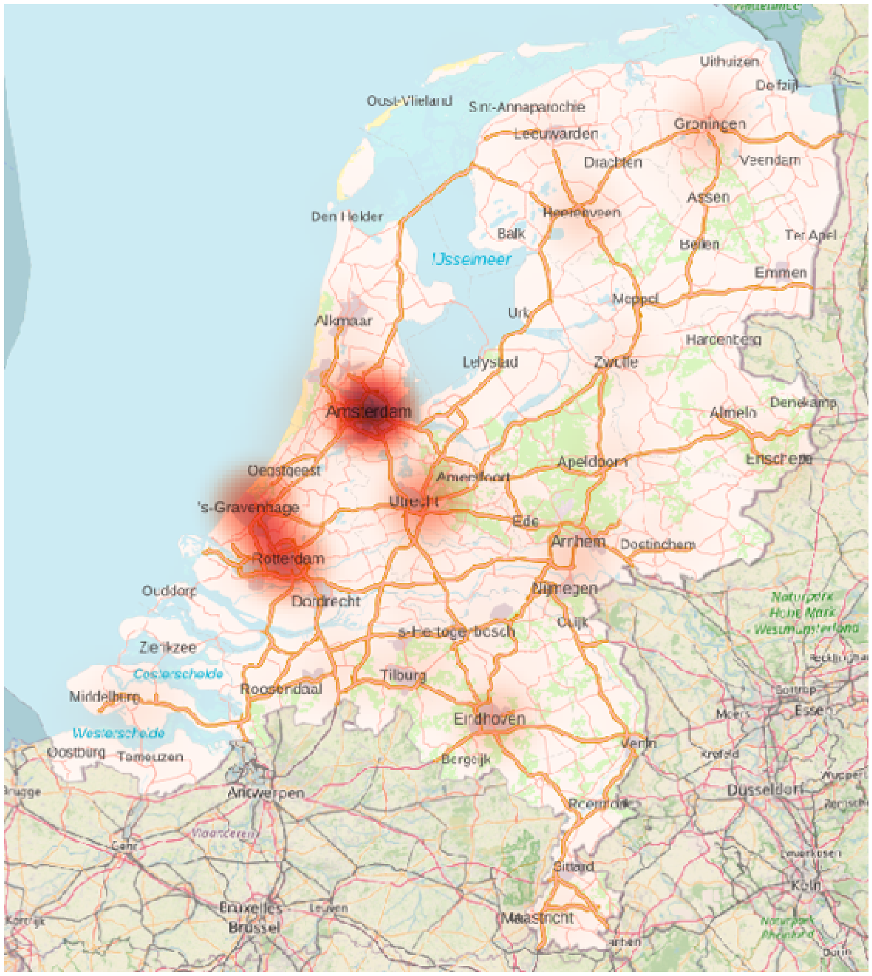
The mean scores show the same general pattern. Bad news increases from a raw mean of 29.1 to a calibrated mean of 45.6, entertainment from 12.6 to 29.6, and celebrity from 2.9 to 11.3. Magnitude decreases from 21.1 to 14.4, while power elite changes only slightly from 16.2 to 17.1. The strong-score columns provide an important qualification: the number and share of strongly scored articles remain unchanged for all six values. The calibration therefore primarily changes the broader presence of news values rather than the set of articles in which a value was already scored as strongly present. This distinction is important when interpreting differences between the raw and calibrated spatial patterns below.

4.2 Spatial distribution of news coverage

The spatial distribution of NU.nl location mentions is highly uneven. The overall location-mention heatmap shows a clear concentration in the Randstad, with Amsterdam as the strongest cluster, followed by The Hague and Rotterdam. Utrecht, Groningen, Eindhoven, and several other urban centres also stand out, but the intensity of location-linked coverage is visibly highest in the west and centre of the country (Figure 4.1). This establishes an important baseline before news values are considered: NU.nl's geographical attention is already spatially concentrated.

Figure 4.1

Overall location mention density in NU.nl coverage

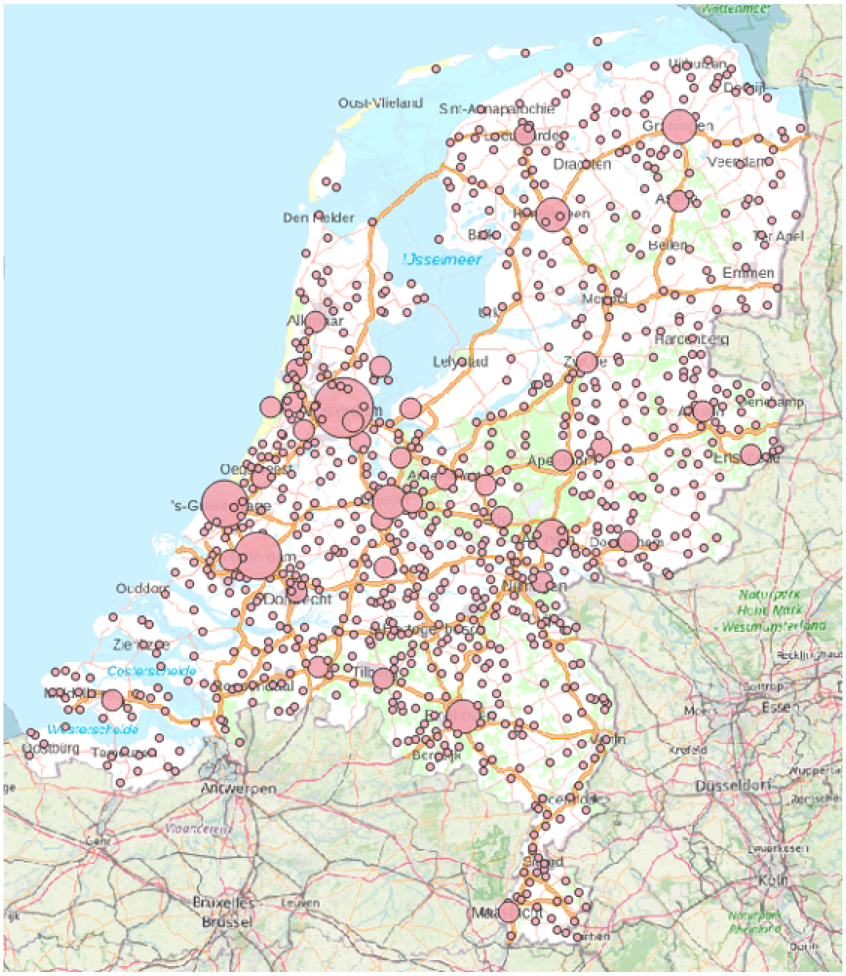


Note. This heatmap visualizes the density of retained Dutch article-location mentions. It shows spatial attention alone.

The mention-count-by-location map confirms this pattern at the level of individual geocoded places (Figure 4.2). Locations throughout the Netherlands appear in the dataset, but the volume of attention differs strongly between places. The largest circles are concentrated around major cities and other nationally visible locations, while many smaller places receive comparatively little coverage. The resulting pattern is therefore national in reach, but strongly unequal in intensity.

Figure 4.2

Mention count by geocoded location



Note. Circles represent unique geocoded locations. Larger circles indicate more article-location mentions.

At the location level, Amsterdam is by far the most frequently mentioned place, followed by Rotterdam and The Hague. Utrecht, Groningen, Eindhoven, Heerenveen, Arnhem, Tilburg, and Breda also appear among the ten most frequently mentioned locations (Table 4.2). The presence of Heerenveen in this list shows that visibility is not determined by population size alone. Smaller or medium-sized places can also become highly visible when they repeatedly occur in nationally covered stories.

Table 4.2

Ten most frequently mentioned locations in the final GIS dataset

Location	Mention count	Dominant raw news value	Dominant calibrated news value
Amsterdam	2,335	Bad news	Bad news
Rotterdam	1,482	Bad news	Bad news
The Hague	1,452	Bad news	Bad news
Utrecht	842	Bad news	Bad news
Groningen	651	Magnitude	Bad news
Eindhoven	610	Bad news	Bad news
Heerenveen	472	Good news	Entertainment
Arnhem	390	Bad news	Bad news
Tilburg	282	Bad news	Bad news
Breda	281	Bad news	Bad news

The raw and calibrated dominance columns in Table 4.2 show that the dominant profile of most highly visible locations remains stable despite calibration. Eight of the ten locations are dominated by bad news in both versions. Groningen and Heerenveen are the exceptions. Groningen changes from magnitude in the raw results to bad news in the calibrated results, while Heerenveen changes from good news to entertainment. Calibration therefore does not affect the volume of attention received by a location, but it can change the type of newsworthiness that appears most prominent after its article-level scores are aggregated.

Location-level checks in the calibrated data further illustrate these differences. Bad news is present in 64.0% of Amsterdam mentions, 65.7% of Rotterdam mentions, and 67.5% of The Hague mentions. Entertainment-oriented visibility is more concentrated in particular places: Heerenveen has an entertainment present share of 83.7%, while Zandvoort has an entertainment present share of 58.3%. These patterns show why frequency and news-value composition should be treated as separate dimensions of geographical news visibility.

A province-level count of article-location mentions provides a broader view of the same spatial concentration (Table 4.3). Zuid-Holland and Noord-Holland together account for more than 40% of all article-location mentions in the municipality aggregation. Noord-Brabant, Gelderland, and Utrecht follow, while Flevoland and Drenthe have the lowest counts. These figures support the visual impression from Figures 4.1 and 4.2: the geography of NU.nl attention extends across the Netherlands, but the volume of coverage is strongly concentrated in particular urban and regional centres.

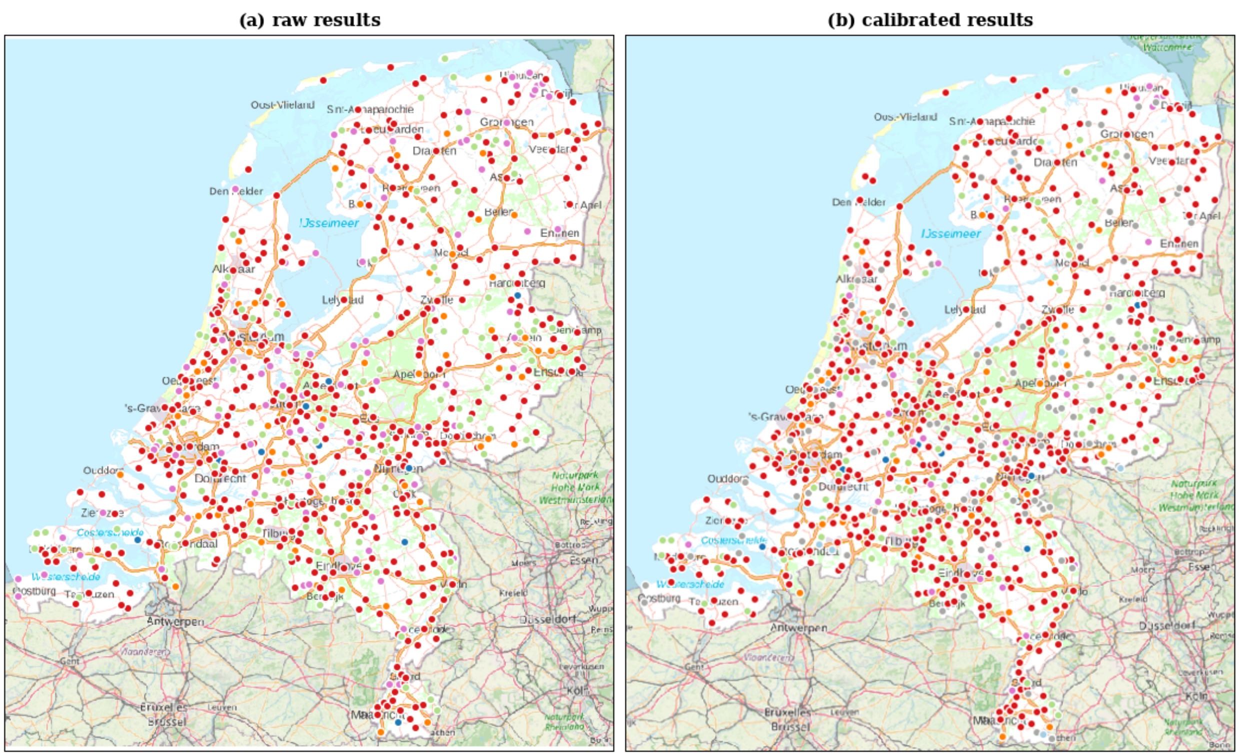
Table 4.3
Article-location mentions by province in the municipality aggregation

Province	Article-location mentions	Share of all mentions
Zuid-Holland	4,235	21.4%
Noord-Holland	4,087	20.7%
Noord-Brabant	2,393	12.1%
Gelderland	2,191	11.1%
Utrecht	1,863	9.4%
Overijssel	1,174	5.9%
Fryslân	939	4.7%
Groningen	935	4.7%
Limburg	733	3.7%
Zeeland	573	2.9%
Drenthe	347	1.8%
Flevoland	304	1.5%

4.3 Spatial distribution of raw and calibrated news values

The dominant-news-value point map adds a first news-value layer to the geography of location mentions (Figure 4.3). Bad news is the most widespread dominant category at point level, although locations dominated by entertainment, magnitude, good news, and power elite are also visible. The point map demonstrates that locations differ not only in how often they are mentioned, but also in the form of newsworthiness most strongly associated with them. Because a large number of individual locations are displayed, however, the map is visually dense. Municipality-level aggregation provides a clearer basis for comparing the raw and calibrated spatial patterns.

Figure 4.3
Dominant news value by geocoded location



Note. Colors are consistent across the thesis: red = bad news; green = entertainment; dark blue = power elite; purple = magnitude; orange = good news; light blue = celebrity.

Table 4.4 compares the raw and calibrated results using mention-weighted municipality aggregation. Each article-location row contributes one observation, meaning that frequently mentioned places have greater influence on the municipal averages. Only municipalities with at least ten article-location mentions are included, resulting in 223 municipalities in each comparison.

Table 4.4

Dominant municipality-level news values using raw and calibrated mention-weighted scores

Scores	Condition	Bad	Ent.	Good	Mag.	Power	Celeb.
Raw	Full dominance, $n \geq 10$ mentions	164	20	16	22	1	0
Calibrated	Full dominance, $n \geq 10$ mentions	188	27	6	2	0	0
Raw	No-bad dominance, $n \geq 10$ mentions	–	20	25	153	23	2
Calibrated	No-bad dominance, $n \geq 10$ mentions	–	111	14	40	56	2

Note. Dominance is based on mention-weighted mean news value scores. The no-bad condition recalculates the dominant news value after excluding bad news from the candidate values. Only municipalities with at least ten article-location mentions are included ($n = 223$).

The full-dominance results show that bad news is already the most common dominant municipal category in the raw data, but becomes substantially more dominant after calibration. In the raw results, bad news dominates 164 of the 223 municipalities. Magnitude is dominant in 22 municipalities, entertainment in 20, good news in 16, and power elite in one, while celebrity is not dominant in any municipality. In the calibrated results, the number of bad-news-dominated municipalities increases to 188. Entertainment is dominant in 27 municipalities, while good news falls to six and magnitude to only two. Neither power elite nor celebrity is dominant in the calibrated full-dominance results.

The contrast between panels (a) and (b) of Figure 4.4 therefore shows that calibration affects the spatial balance between the categories. The raw map contains substantially more magnitude- and good-news-dominated municipalities, whereas the calibrated map is more strongly dominated by bad news. This geographical shift corresponds to the corpus-level comparison in Table 4.1, where the prevalence of magnitude decreases after calibration while bad news becomes substantially more common.

The no-bad-news results make the effect of calibration even clearer. These calculations do not remove bad-news articles from the data; they exclude bad news only from the set of candidate values when determining which news value has the highest municipal mean score. In the raw no-bad results, magnitude overwhelmingly dominates the secondary geography, appearing as the strongest remaining value in 153 of the 223 municipalities. Good news is dominant in 25 municipalities, power elite in 23, entertainment in 20, and celebrity in two.

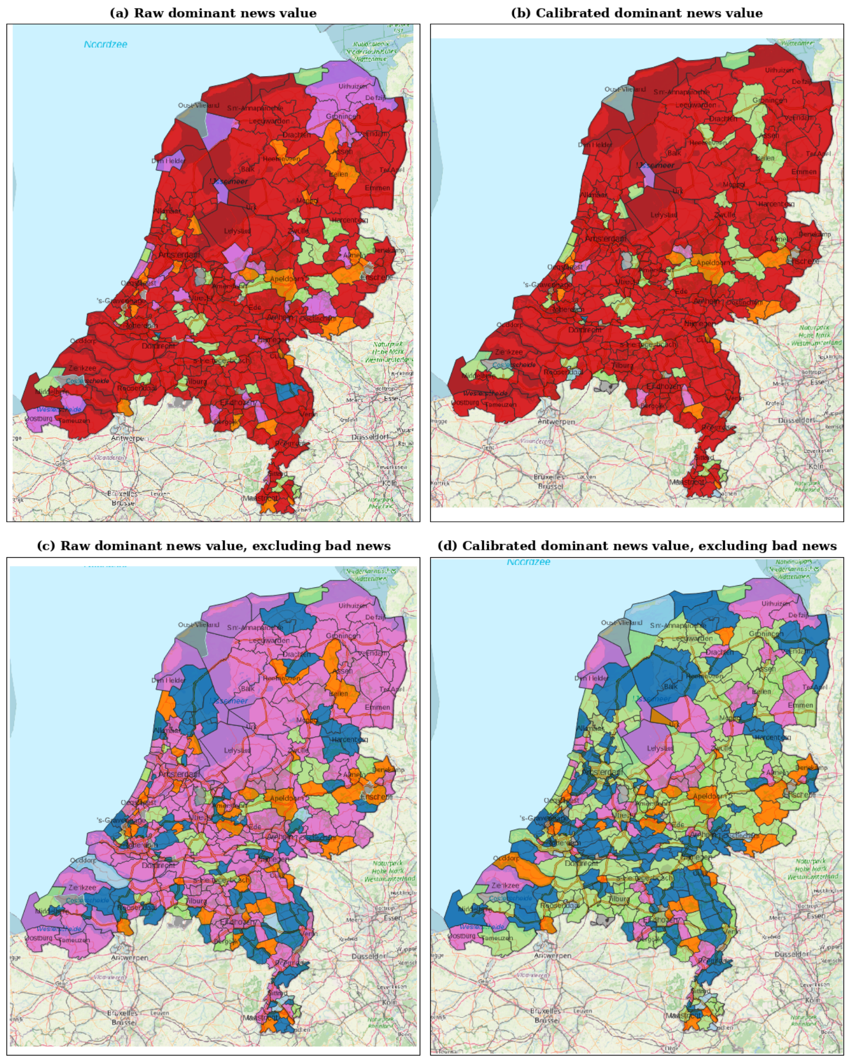
After calibration, this pattern changes substantially. Entertainment becomes the most common secondary dominant value, increasing from 20 municipalities in the raw results to 111 in the calibrated results. Power elite also becomes more prominent, increasing from 23 to 56 municipalities. Magnitude falls sharply from 153 to 40 municipalities, while good news decreases from 25 to 14. Celebrity remains dominant in two municipalities.

Panels (c) and (d) of Figure 4.4 therefore provide a particularly clear spatial comparison between the raw and calibrated classifications. When bad news is excluded from the dominance calculation, the raw geography is largely organized around magnitude. In the calibrated results, the secondary geography becomes considerably more varied and is primarily structured around entertainment, power elite, and magnitude. This is consistent with the corpus-level changes shown in Table 4.1, where calibration reduces the prevalence of magnitude while substantially increasing entertainment.

Taken together, the municipality-level results show that calibration affects not only corpus-level prevalence but also the geographical interpretation of the news values. Bad news becomes more spatially dominant in the full calibrated results, while the most common secondary dominant value changes from magnitude in the raw results to entertainment after calibration. The calibrated spatial patterns should therefore not be understood as a simple rescaling of the raw scores. Instead, calibration changes the relative balance between news values in ways that are also visible geographically.

Figure 4.4

Dominant news value by municipality



Note. All panels show municipality-level dominant news values using mention-weighted aggregation and include only municipalities with at least ten article-location mentions. Panel (a) shows dominance based on the raw scores, while panel (b) shows dominance based on the calibrated scores. Panels (c) and (d) repeat these calculations after excluding bad news from the dominance calculation, using the raw and calibrated scores respectively, in order to reveal secondary news value patterns.

At province level, aggregation smooths much of the municipality-level variation, but the comparison between raw and calibrated scores still reveals a clear effect of calibration (Figure 4.5). In the raw full-dominance results, bad news is dominant in 11 of the 12 provinces. Groningen is the only exception, where magnitude narrowly exceeds bad news: the raw mean magnitude score is 28.6 compared with a bad-news mean of 28.4. In the calibrated results, bad news again dominates 11 provinces, but the exception changes. Groningen becomes bad-news dominant, while Fryslân becomes entertainment dominant. The full-dominance maps therefore show that bad news remains the main province-level category in both versions, but calibration changes which regional exception remains visible.

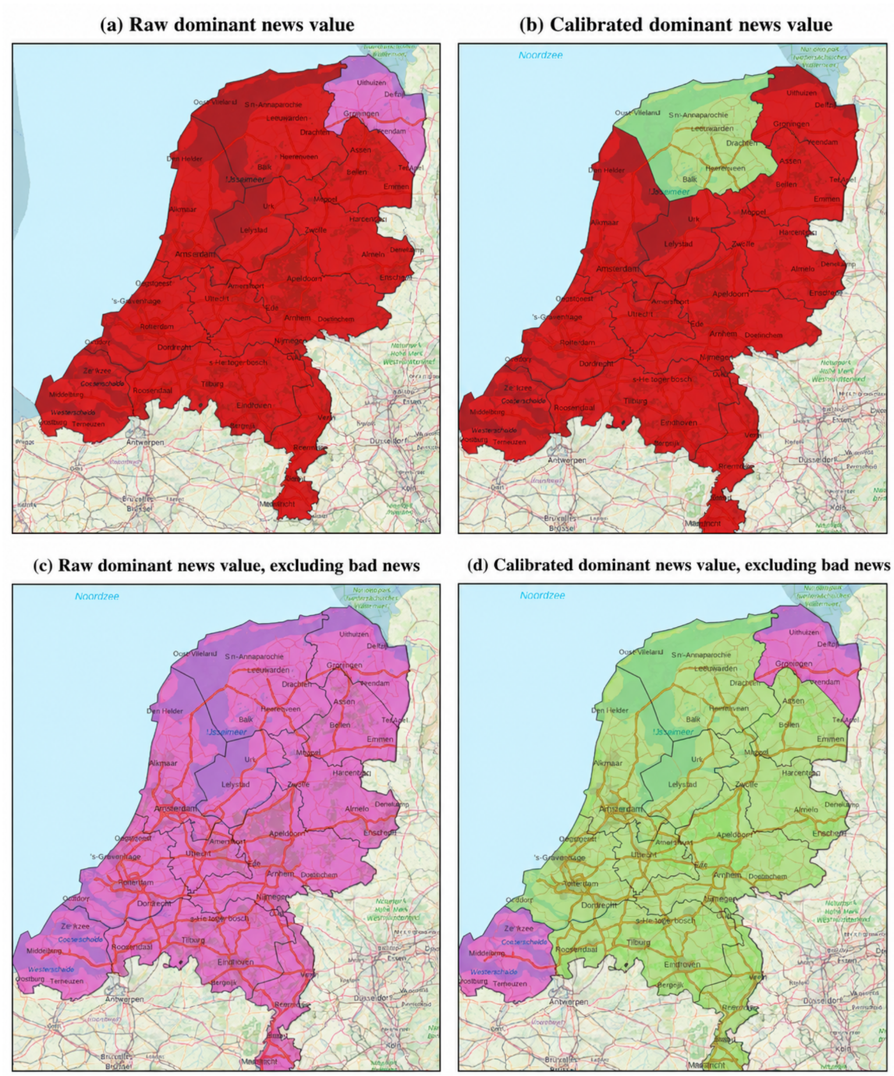
The difference between the raw and calibrated results becomes much stronger when bad news is excluded from the dominance calculation. In the raw results, magnitude is the dominant remaining news value in all 12 provinces. This produces an almost completely uniform province-level pattern in panel (c). The result is consistent with the corpus-level raw scores, in which magnitude is substantially more prevalent than entertainment and other non-bad-news values.

After calibration, the secondary province-level geography changes considerably. Entertainment becomes the dominant non-bad-news value in 10 of the 12 provinces, while Groningen and Zeeland remain magnitude dominant. Panel (d) therefore contrasts sharply with the raw no-bad map: a pattern that is uniformly magnitude-oriented before calibration becomes predominantly entertainment-oriented after calibration, with only two magnitude-oriented regional exceptions. This shift reflects the broader effect of calibration observed throughout the analysis, particularly the reduction of magnitude and the increase of entertainment in the calibrated scores.

The province-level results also show the effect of spatial scale. Municipality-level maps retain substantial local variation, whereas province aggregation produces much more homogeneous patterns. Nevertheless, comparing raw and calibrated scores at this broader scale remains informative. In the full-dominance maps, the main effect is a change in the exceptional province, from Groningen in the raw results to Fryslân in the calibrated results. In the no-bad maps, the effect is much larger: calibration changes the dominant secondary pattern from magnitude across the entire country to entertainment across most provinces. The raw and calibrated province maps therefore reinforce the conclusion that calibration changes not only overall news-value prevalence, but also the geographical patterns that emerge after aggregation.

Figure 4.5

Dominant news value by province for raw and calibrated scores under full and no-bad-news aggregation



Note. All panels show province-level dominant news values using mention-weighted aggregation. Panel (a) shows dominance based on the raw scores, while panel (b) shows dominance based on the calibrated scores. Panels (c) and (d) repeat these calculations after excluding bad news from the dominance calculation, using the raw and calibrated scores respectively.

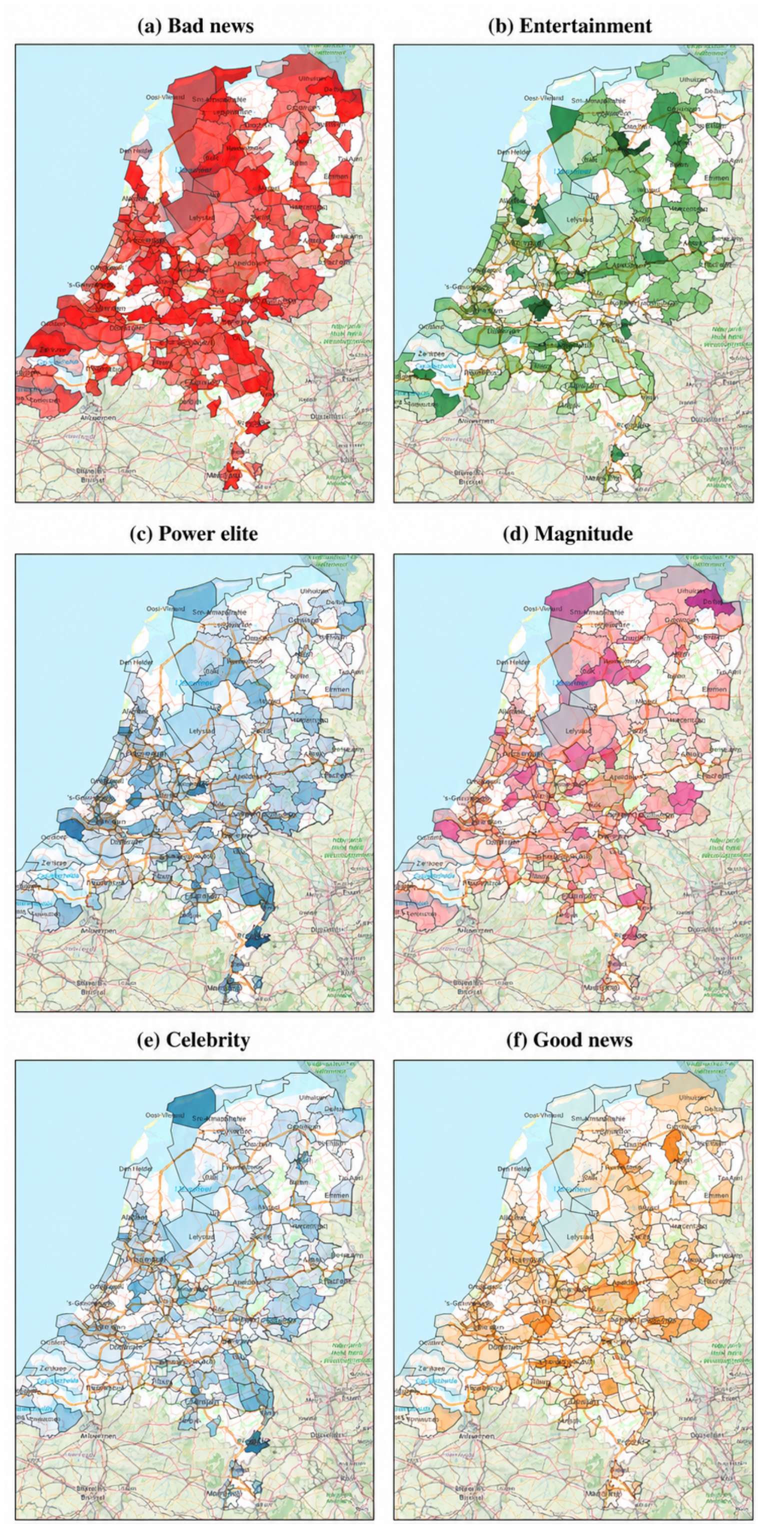
4.4 Relative composition of municipal news values

The present-share maps move beyond dominance by showing how often each news value is present in a municipality's article-location mentions. Rather than assigning each municipality a single category, they show the relative frequency of all six news values independently. This is important because multiple news values can be present within the same municipality even when one value dominates its average profile. Figures 4.6 and 4.7 therefore provide a more detailed comparison of the calibrated and raw spatial distributions. Darker shading indicates a higher present share.

The calibrated present-share maps show that several news values are geographically widespread (Figure 4.6). Bad news has the strongest overall presence and reaches relatively high shares across much of the country. Entertainment is also widely distributed after calibration, although the strongest concentrations remain more spatially selective. Power elite has a broader but generally moderate pattern, while celebrity and good news are present at lower levels but remain visible across a substantial number of municipalities. Magnitude is also present across the country, but its calibrated distribution is less pronounced than in the raw results.

Figure 4.6

Municipality-level present shares of calibrated news value scores



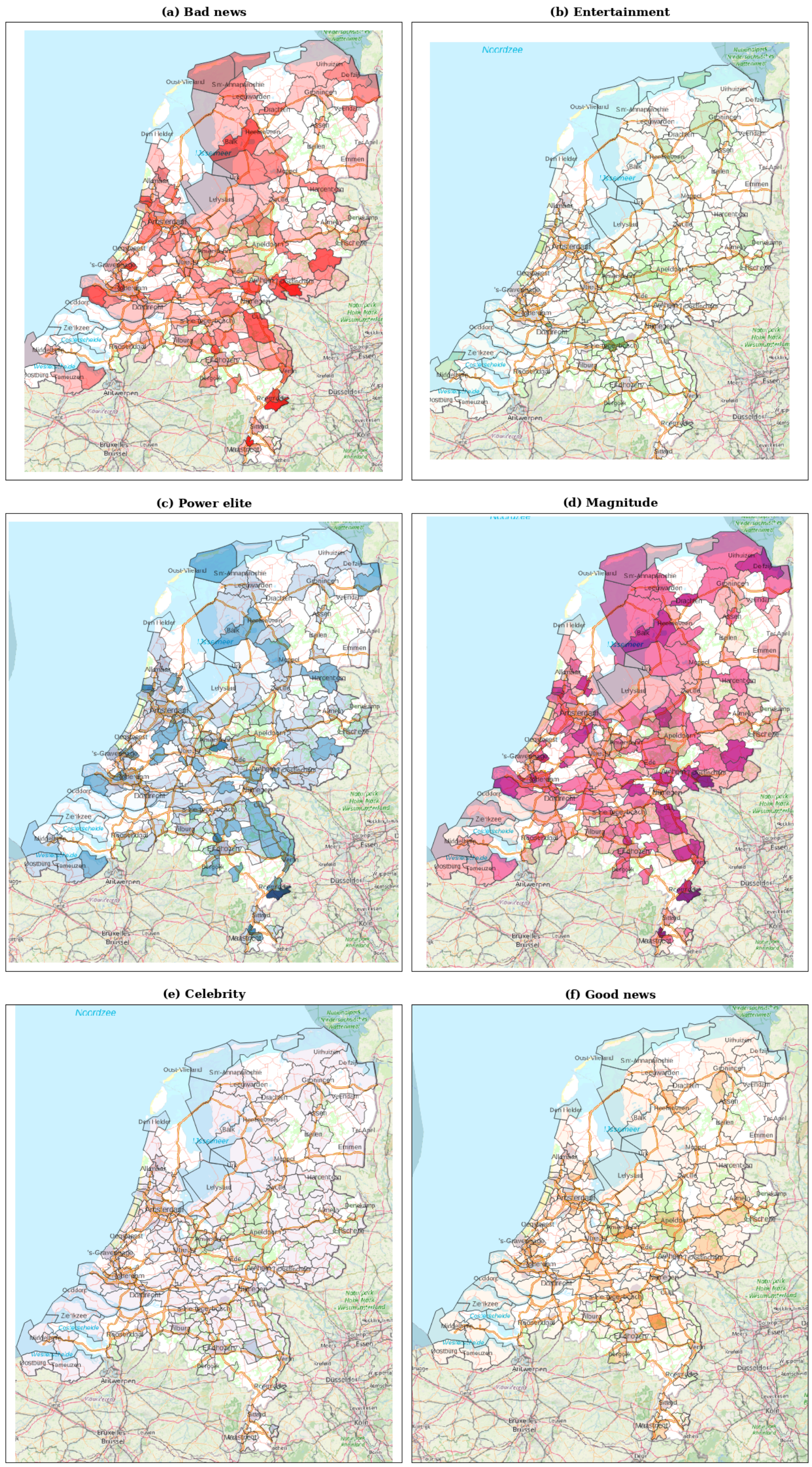
Note. Present share is the proportion of article-location mentions in which the news value received a score above zero. Only municipalities with at least ten article-location mentions are shown. Darker shading indicates a higher present share.

The raw present-share maps are generally lighter than the calibrated maps (Figure 4.7). This reflects the fact that the raw model assigned positive news value scores less frequently for most categories, leaving more article observations with no detected presence for a particular news value and also more articles with no news value assigned at all. As a result, the municipality-level present shares are lower for most news values before calibration. This difference is especially clear for bad news, entertainment, celebrity, and good news. Power elite changes comparatively little between the two versions.

Magnitude is the main exception to this broader pattern. In the raw results, magnitude is considerably more prominent and geographically widespread than after calibration. Many municipalities therefore appear darker for magnitude in the raw map even though the other raw news-value maps are generally lighter. This is consistent with the corpus-level results, where magnitude has a raw present share of 36.6% but a calibrated present share of 23.2%, while the other five news values are either more prevalent or only slightly more prevalent after calibration.

Figure 4.7

Municipality-level present shares of raw news value scores



Note. Present share is the proportion of article-location mentions in which the news value received a score above zero. Only municipalities with at least ten article-location mentions are shown. Darker shading indicates a higher present share.

The contrast between the two figures therefore shows that calibration changes not only which news value becomes dominant, but also how frequently the individual values are detected across municipalities. Bad news becomes substantially more prevalent after calibration, producing darker calibrated shares across much of the country. Entertainment shows an even more pronounced change: its raw distribution is comparatively sparse, whereas the calibrated map shows much broader municipal presence. Celebrity also changes strongly, moving from very limited raw presence to a substantially more visible calibrated pattern. Good news becomes somewhat more widespread, while power elite remains comparatively stable.

Magnitude moves in the opposite direction. In the raw results, it is one of the most geographically prominent news values and reaches relatively high present shares in many municipalities. After calibration, its overall presence is reduced and the strongest areas become more selective. This corresponds closely to the earlier dominance results, where magnitude dominates 153 municipalities in the raw no-bad calculation but only 40 after calibration. The present-share maps therefore show that this difference is not merely an artefact of selecting a single dominant category: the underlying frequency of magnitude is itself substantially higher in the raw results.

Taken together, Figures 4.6 and 4.7 show that the raw classifications generally produce lower present shares and therefore lighter municipal maps because the model assigns fewer positive news value scores. Calibration increases the detected presence of five of the six values, most clearly bad news, entertainment, celebrity, and good news, while magnitude decreases. Power elite changes only slightly. These spatial differences closely reflect the corpus-level comparison in Table 4.1 and demonstrate that calibration affects not only aggregate prevalence but also the geographical distribution of news value presence across Dutch municipalities.

4.5 Discussion

The results show that NU.nl coverage produces a differentiated geography of Dutch places. The first and most visible pattern is spatial unevenness. Large cities and institutionally visible places receive the most location-linked attention, especially Amsterdam, Rotterdam, The Hague, Utrecht, Groningen, and Eindhoven. This finding is consistent with previous research on spatial bias and geographic diversity in news, which shows that news coverage is geographically patterned rather than randomly distributed (Özgün, 2023; Vogler et al., 2023). It also fits the broader media-geographical argument that journalism does not merely report on places, but helps organize the visibility and meaning of places (Adams & Jansson, 2012; Gutsche & Hess, 2019; Usher, 2020).

The second pattern is that locations are not merely mentioned more or less often, but are made visible through different forms of newsworthiness. In the calibrated results, bad news is the strongest overall pattern and dominates a large proportion of the municipality- and province-level geography. This is consistent with news value theory, where negativity or bad news has long been treated as a central criterion of newsworthiness (Harcup & O'Neill, 2017). The spatial analysis adds a geographical dimension to this observation by showing where such forms of newsworthiness become attached. Major cities are therefore not only prominent in terms of coverage volume, but are also repeatedly visible through articles involving harm, risk, crisis, crime, public problems, or other negative consequences.

At the same time, the raw and calibrated comparisons show that this geography is sensitive to the classification procedure. The raw scores produce substantially less bad-news and entertainment presence and substantially more magnitude presence. This difference is especially visible in the no-bad-news dominance maps. Before calibration, magnitude is the dominant remaining news value in 153 of the 223 municipalities and in all twelve provinces. After calibration, entertainment becomes the dominant secondary value in 111 municipalities and in ten of the twelve provinces. The present-share maps confirm that these differences are not only caused by assigning a single dominant category: magnitude is geographically much more prevalent in the raw scores, whereas entertainment becomes much more widespread after calibration.

The comparison therefore provides an important methodological qualification to the substantive results. The calibrated scores performed better against the manually coded validation sample and are consequently treated as the more reliable analytical version, but they should not be understood as an objective correction that reveals a single true geography of newsworthiness. Rather, the raw and calibrated maps together show how model performance and calibration assumptions influence the spatial patterns that become visible. The raw results are particularly useful as a diagnostic comparison because they demonstrate the consequences of the model's under-detection of entertainment and celebrity and over-detection of magnitude.

The maps also show that bad news is not the only route through which places become visible. Entertainment, power elite, magnitude, good news, and celebrity remain geographically differentiated even when they do not dominate the full calibrated maps. The present-share maps are especially important in this respect because they show that a municipality can contain substantial levels of several news values even when bad news has the highest mean score. Entertainment-oriented visibility is particularly apparent in selected municipalities associated with sport, culture, events, and leisure, while power-elite patterns reflect more institutional forms of visibility. Magnitude remains geographically important in particular areas even after calibration, although its overall prevalence is lower than in the raw output.

Spatial scale also affects interpretation. Municipality-level aggregation preserves considerably more local variation, whereas province-level aggregation smooths these differences and produces more homogeneous patterns. In the full province maps, bad news dominates eleven of the twelve provinces in both the raw and calibrated versions, although the exceptional province changes from Groningen in the raw results to Fryslân in the calibrated results. The no-bad province comparison shows a stronger calibration effect, changing from magnitude dominance across all provinces in the raw results to entertainment dominance across most provinces after calibration. The geography of newsworthiness should therefore be understood as a pattern that becomes more or less differentiated depending on both classification and spatial aggregation.

Taken together, these findings extend media-geographical research by showing that spatial bias is not only a matter of which places are mentioned. It also concerns the forms of newsworthiness through which those places become visible. The combination of location frequency, news value classification, raw-versus-calibrated comparison, and spatial aggregation therefore provides a way of examining both the distribution and the character of mediated geographical attention.

4.6 Reliability and limitations

The results should be interpreted as patterns in representation rather than as direct measures of local reality. A high bad-news score for a municipality does not mean that the municipality itself is more dangerous or negative. It means that, in this NU.nl corpus, the municipality appears relatively often in articles where bad news contributes to newsworthiness. This distinction follows from the theoretical approach of the thesis: news values are treated as discursive traces in published news output, not as direct reconstructions of editorial decision-making or of social conditions in places (Bednarek & Caple, 2017; Harcup & O'Neill, 2017; Mast & Temmerman, 2021).

The first limitation concerns news value assignment. The LLM-assisted classification makes large-scale analysis possible, but it remains interpretive. The model applies a predefined codebook to article excerpts and does not measure newsworthiness objectively. The validation showed that the calibrated output performs better overall than the raw output, but it is not precise enough for strong claims about every individual article-location pair. Bad news had high recall but lower precision, meaning that it may be over-detected after calibration. Good news had the weakest F1-score and is therefore interpreted most cautiously. This is consistent with broader work on computational and LLM-assisted annotation, which stresses that such methods require clear codebooks, validation, and error analysis rather than being treated as neutral measurement tools (Cheema et al., 2025; Gilardi et al., 2023; Törnberg, 2024; Ziems et al., 2024).

A second limitation concerns benchmark calibration. Calibration improved the overall validation results and corrected important weaknesses in the raw model output, particularly the under-detection of entertainment and celebrity and the over-detection of magnitude. However, it also introduces a modelling assumption. The calibrated distribution is adjusted toward benchmark prevalence derived from Harcup and O'Neill's news value data. This does not establish that NU.nl's 2025 digital coverage naturally follows the same distribution. This domain difference is substantial because the benchmark is based on newspaper page-lead stories, whereas the present study analyzes a continuously published digital-news corpus. The calibrated scores should therefore be understood as adjusted indicators rather than as direct observations of an objectively correct prevalence.

The direct comparison between raw and calibrated GIS results makes this limitation particularly visible. Calibration changes not only corpus-level frequencies but also the resulting geographical patterns. For example, magnitude dominates the raw no-bad-news municipality and province maps, whereas entertainment becomes substantially more prominent after calibration. The calibrated maps are supported by the stronger validation performance, but these differences demonstrate that spatial interpretation remains partly dependent on the calibration procedure. Presenting both versions therefore improves methodological transparency but does not eliminate the underlying uncertainty.

The third limitation concerns the unit of analysis. News values were classified at article level, after which all locations in an article inherited the same scores. This is methodologically practical and theoretically defensible because news values often operate at the article level, but it can over-attach news values to incidental locations. If an article primarily concerns one place but briefly mentions another, both locations receive the same article-level scores. The maps therefore show where locations appear in articles with certain news values, not necessarily where those news values physically occur or apply equally to every mentioned place.

The fourth limitation concerns excerpt-based classification. Articles were classified using short excerpts rather than full texts. This made the workflow feasible for a large corpus, but news values that appear later in an article may be missed or receive less weight than features appearing in the lead. The validation evaluates the actual excerpt-based workflow used in the thesis, so this limitation is already partly reflected in the performance metrics, but it remains important for interpretation.

The fifth limitation concerns location extraction and geocoding. Place-name recognition and coordinate matching are imperfect steps in spatial text analysis. Some place names may be missed, ambiguous names may be resolved incorrectly, administrative areas may be represented by a single coordinate, and some locations may be mentioned only incidentally. This is a known issue in spatial humanities and geolocated news analysis (Kriesch & Losacker, 2025; Peris et al., 2020; Won et al., 2018). The maps should therefore be treated as analytical representations of extracted and geocoded NU.nl location mentions, not as complete or perfectly precise maps of all places involved in the news.

Representing administrative names as points is particularly important for municipality-level aggregation. A reference to a province such as Zeeland can be represented by one coordinate and consequently assigned to one municipality during a spatial join, even though the textual reference concerns a much larger geographical area. Such cases can disproportionately affect the aggregated profile of the municipality containing that point. This does not invalidate the broader national patterns, but it means that individual municipal results should be interpreted cautiously when they are strongly influenced by ambiguous or administrative place names.

The sixth limitation concerns GIS aggregation and visualization. The municipality and province maps use mention-weighted aggregation, meaning that every article-location row contributes one observation. Frequently mentioned locations therefore have greater influence on the aggregated results than locations appearing only occasionally. This is appropriate for analyzing the geography of NU.nl coverage volume, but it means that the maps do not represent all unique places equally. Province-level aggregation additionally smooths much of the municipality-level variation, so broad regional patterns can obscure important local differences.

Dominance maps also simplify a multidimensional pattern by assigning one news value to each spatial unit. Where two or more mean scores are close, the identity of the highest category can depend on relatively small differences. The no-bad-news maps should therefore be understood as secondary analytical views rather than replacements for the full dominance maps. They deliberately remove bad news from the set of candidate categories in order to reveal which other value is strongest, but this does not mean that bad news is absent from those municipalities or provinces.

The present-share maps reduce some of this simplification by displaying each news value independently, but they also become unstable when based on small numbers of observations. For this reason, the main municipality-level dominance and present-share visualizations include only municipalities with at least ten article-location mentions. The same threshold is applied to the raw and calibrated results so that the comparisons are based on the same set of municipalities. This improves comparability and reduces extreme percentages based on very small samples, but the threshold remains an analytical choice that shapes which municipalities appear in the main maps.

A further technical limitation concerns failed model-output parsing. After repeated parsing or validation failures, the original workflow stored fallback zero scores so that failed cases remained visible in the output. This affected 147 classified articles, corresponding to 186 article-location rows in the GIS dataset. A technical parsing failure is not substantive evidence that all six news values are absent, so these fallback values may introduce a small downward bias in the raw classifications and may also affect the subsequent calibration ranking. Because the overall parse-success rate was 98.8%, the affected share is small, but these cases should nevertheless be interpreted as technical failures rather than genuine all-zero classifications.

A final limitation concerns the scope of the news value taxonomy. The thesis focuses on six selected news values: entertainment, bad news, magnitude, good news, celebrity, and power elite. Other values from Harcup and O'Neill's taxonomy, such as conflict, relevance, follow-up, surprise, exclusivity, shareability, audio-visuals, drama, and the news organization's agenda, are not operationalized. This narrowing was necessary for feasibility and reliability, because several of these values require editorial context, visual material, platform metrics, or cross-article comparison that was not available in the extracted text fields. However, it also means that the analysis captures only part of the broader news value taxonomy.

These limitations mean that the strongest claims in this thesis concern aggregate spatial patterns rather than individual articles, individual locations, or the full complexity of newsworthiness. The calibrated results are preferred because they perform better in manual validation, but the raw-versus-calibrated comparison demonstrates that calibration materially influences some of the observed spatial patterns. Bad news is the dominant calibrated form of spatial visibility, while entertainment becomes the most prominent secondary value in many calibrated areas. Magnitude is much more prominent in the raw results, illustrating the importance of treating model output and calibration critically. Good news remains the least reliable category, while power elite changes comparatively little between the raw and calibrated distributions. The results are therefore most reliable when interpreted as broad patterns of mediated place representation rather than as precise classifications of individual municipalities.

5 Conclusion

This thesis asked how news values are geographically distributed in Dutch digital journalism. The results show that this distribution is uneven in two connected ways. First, NU.nl's location-linked attention is concentrated in major urban and institutionally visible places, especially Amsterdam, Rotterdam, The Hague, Utrecht, Groningen, and Eindhoven. Second, the forms of newsworthiness attached to Dutch places vary geographically. These patterns are visible at individual location, municipality, and province level, while the comparison between raw and calibrated scores also shows that the observed geography depends partly on how news values are computationally classified and adjusted.

The strongest calibrated pattern is the dominance of bad news. Bad news is the dominant category in 188 of the 223 municipalities included in the main mention-weighted analysis and in eleven of the twelve provinces. This should not be interpreted as evidence that these places are themselves more negative or dangerous. It shows that they are frequently mentioned in NU.nl articles in which harm, danger, crisis, crime, loss, or other negative consequences contribute to newsworthiness. The spatial analysis therefore demonstrates how a major news value becomes geographically attached to places through repeated news representation.

The raw-versus-calibrated comparison is an important part of this conclusion. Bad news is already the most common dominant municipal category in the raw results, but calibration strengthens this pattern. More importantly, the secondary geography changes considerably. When bad news is excluded from the dominance calculation, magnitude dominates 153 of 223 municipalities in the raw results and all twelve provinces. In the calibrated results, entertainment instead becomes dominant in 111 municipalities and ten of the twelve provinces, while magnitude becomes much less prevalent. The present-share maps show the same broader change: calibration increases the geographical prevalence of bad news, entertainment, celebrity, and good news, reduces magnitude, and leaves power elite comparatively stable.

These differences demonstrate why validation and methodological transparency are important when LLM-assisted classifications are used for spatial analysis. The calibrated version performed better against the manually coded sample and is therefore treated as the preferred analytical representation. At the same time, comparing it with the raw model output shows that calibration does more than adjust numerical prevalence: it can alter the spatial patterns that emerge after aggregation. The maps should consequently be interpreted as analytical representations produced through a sequence of classification, calibration, geocoding, aggregation, and visualization decisions.

Despite this qualification, the analysis shows that places are not made visible through one form of newsworthiness alone. Entertainment, magnitude, good news, celebrity, and power elite produce distinct spatial patterns alongside bad news. The present-share maps make this especially clear because they show that several values can be substantially present within the same municipality even when only one becomes dominant. News visibility is therefore both quantitative and qualitative: places differ in how often they appear and in the forms of newsworthiness through which they are repeatedly represented.

The thesis contributes to digital humanities by combining computational text analysis, LLM-assisted annotation, manual validation, geocoding, and GIS mapping in one transparent workflow. It demonstrates that news values can be treated as spatially analyzable features of published digital news and that comparing raw and calibrated classifications can make the methodological sensitivity of such analyses visible. It also contributes to media-geographical research by connecting uneven spatial attention to the particular forms of newsworthiness through which places enter national digital coverage.

The main limitations are the use of article-level rather than location-specific news value scores, excerpt-based classification, assumptions introduced through benchmark calibration, possible errors in location extraction and geocoding, the influence of frequently mentioned places in mention-weighted aggregation, and the simplification involved in assigning dominant categories to spatial units. Future research could address these limitations through sentence- or paragraph-level classification, full-text annotation, alternative or corpus-specific calibration strategies, comparison with other Dutch news outlets, and longitudinal analysis of how the news value profiles of places change over time. Overall, the thesis demonstrates that mapping news values makes visible a dimension of digital journalism that is usually less explicit: not only where news attention is directed, but how places become newsworthy.

A Research assistance using ChatGPT

ChatGPT was used as coding and writing support during the development of the Python workflow and thesis text. It supported implementation tasks such as intermediate batch saving and resume-safe processing, article-ID matching and join logic, prompt construction, JSON parsing and retry logic, model-loading code, benchmark calibration code, GIS table rebuilding, aggregation logic, validation metric functions, and the reshaping of diagnostic comparison files. ChatGPT was also used to help revise wording and structure in parts of the thesis text. It was not used as the article-level news value classifier: those classifications were produced using Qwen2. All substantive decisions about the research design, news value definitions, manual validation, interpretation of results, and final thesis text were checked and approved by the author.

B Qwen2 news value classification prompt

```
SYSTEM PROMPT

Apply a deductive news value codebook to Dutch NU.nl article excerpts.

Classify each article at article level using six independent scores: 0, 50, or 100.

News values to classify:

1. entertainment_score
Soft or engaging newsworthiness: sport, culture, media, showbusiness,
lifestyle, leisure, animals, festivals, humour, spectacle, or lighter
human interest.

2. bad_news_score
Negative newsworthiness: harm, death, injury, illness, danger, loss,
damage, crime, punishment, crisis, failure, threat, or serious negative
consequences.

3. magnitude_score
Scale or impact: large numbers, many people affected, major financial
amounts, broad consequences, national or regional impact, records,
exceptional scale, or extreme occurrence.

4. good_news_score
Positive newsworthiness: success, victory, achievement, recovery, rescue,
improvement, celebration, breakthrough, positive return, favourable
outcome, or a problem being solved.
```

5. celebrity_score

Famous people: well-known athletes, royals, artists, presenters, influencers, entertainers, or other public figures whose public profile contributes to the story.

6. power_elite_score

Powerful actors or institutions: government, police, courts, ministries, municipalities, regulators, major companies, public authorities, hospitals, universities, or other institutions when they are central actors.

Scoring:

0 = not meaningfully present.

50 = present but secondary.

100 = central to why the article is newsworthy.

Multi-label rule:

Evaluate all six values independently. Several values may be present in the same article.

dominant_news_values:

Return the value or values with the highest non-zero score.

Use ["none"] only if all six scores are 0.

Scores must be assigned independently for each news value. Celebrity and good news should not be treated as exceptional categories. Code celebrity when an already famous person contributes to the article's newsworthiness. Code good news when positive outcomes such as wins, recoveries, rescues, breakthroughs, celebrations, or successful interventions are present, including when negative elements are also present.

Return only valid JSON with these fields:

- entertainment_score
- bad_news_score
- magnitude_score
- good_news_score
- celebrity_score
- power_elite_score
- dominant_news_values
- short_reason

Use this structure:

```
{
  "entertainment_score": 0,
  "bad_news_score": 100,
  "magnitude_score": 50,
  "good_news_score": 0,
  "celebrity_score": 0,
  "power_elite_score": 50,
  "dominant_news_values": ["bad_news"],
  "short_reason": "Negative consequences are central; institutional actors are secondary."
}
```

USER PROMPT

Classify this NU.nl article excerpt according to the codebook.

ARTICLE TO CLASSIFY:

```
Article ID: [article ID]

Date: [publication date]

Title: [article title]

Subjects: [Nexis subject metadata]

Detected Dutch locations: [detected locations]

Text source used: [lead_text / lead_text + body_text / body_text]

Article excerpt: [standardized article excerpt]


Return only the JSON object.
```

C Few-shot examples

The final Qwen2 classification run included four manually coded few-shot examples. These examples were included in the prompt context to demonstrate how the operational definitions and the three-level scoring scheme should be applied. They were the first four rows of the manually coded few-shot example file used in the classification workflow.

Example 1: Article 8741

```
Title: Deze man waakte dag en nacht over onze veiligheid
Detected Dutch locations: Amsterdam; Den Haag; Rotterdam; Utrecht; Zwolle

Expected scores:
entertainment_score: 50
bad_news_score: 0
magnitude_score: 0
good_news_score: 0
celebrity_score: 0
power_elite_score: 100
```

Example 2: Article 39258

```
Title: Politie vermoedt 'huiselijk geweld' na dood vrouw door steekpartij in Lent
Detected Dutch locations: Lent; Nijmegen

Expected scores:
entertainment_score: 0
bad_news_score: 100
magnitude_score: 50
good_news_score: 0
celebrity_score: 0
```

power_elite_score: 50

Example 3: Article 7089

Title: Cijfers bij sensatie PSV in Liverpool: eerste Nederlandse zege op Anfield ooit

Detected Dutch locations: Eindhoven

Expected scores:

entertainment_score: 100

bad_news_score: 0

magnitude_score: 50

good_news_score: 100

celebrity_score: 50

power_elite_score: 0

Example 4: Article 54720

Title: Zeldzame bergbongo's arriveren in Nederlandse dierentuin

Detected Dutch locations: Kerkrade

Expected scores:

entertainment_score: 100

bad_news_score: 0

magnitude_score: 0

good_news_score: 50

celebrity_score: 0

power_elite_score: 0

References

Adams, P. C., & Jansson, A. (2012). Communication geography: A bridge between disciplines. *Communication Theory*, *22*(3), 299–318. https://doi.org/10.1111/j.1468-2885.2012.01406.x

Bednarek, M., & Caple, H. (2017). *The discourse of news values: How news organizations create newsworthiness*. Oxford University Press.

Burggraaff, C., & Trilling, D. (2020). Through a different gate: An automated content analysis of how online news and print news differ. *Journalism*, *21*(1), 112–129. https://doi.org/10.1177/1464884917716699

Cheema, G. S., Azimi, M., Ewerth, R., & Müller-Budack, E. (2025). Leveraging large language models for news values analysis. *Disinformation and Misinformation in the Age of Generative AI & Investigating Learning during Web Search 2025*, *4070*, 47–50. https://ceur-ws.org/Vol-4070/short3.pdf

di Buono, M. P., Šnajder, J., Dalbelo Bašić, B., Glavaš, G., Tutek, M., & Milic-Frayling, N. (2017). Predicting news values from headline text and emotions. *Proceedings of the 2017 EMNLP Workshop: Natural Language Processing Meets Journalism*, 1–6. https://doi.org/10.18653/v1/W17-4201

Galtung, J., & Ruge, M. H. (1965). The structure of foreign news: The presentation of the Congo, Cuba and Cyprus crises in four Norwegian newspapers. *Journal of Peace Research*, *2*(1), 64–90. https://doi.org/10.1177/002234336500200104

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences of the United States of America*, *120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

Gutsche, R. E. J., & Hess, K. (2019). *Geographies of journalism: The imaginative power of place in making digital news*. Routledge. https://doi.org/10.4324/9781315148946

Harcup, T., & O'Neill, D. (2017). What is news? news values revisited again. *Journalism Studies*, *18*(12), 1470–1488. https://doi.org/10.1080/1461670X.2016.1150193

Kriesch, L., & Losacker, S. (2025). A geolocated dataset of German news articles. *Scientific Data*, *12*, Article 1128. https://doi.org/10.1038/s41597-025-05422-w

Mast, J., & Temmerman, M. (2021). What's the news? reassessing "news values" as a concept and methodology in the digital age. *Journalism Studies*, *22*(6), 689–701. https://doi.org/10.1080/1461670X.2021.1917445

O'Neill, D., & Harcup, T. (2020). News values and news selection. In K. Wahl-Jorgensen & T. Hanitzsch (Eds.), *The handbook of journalism studies* (2nd ed., pp. 213–228). Routledge.

Özgün, B. (2023). *Geography of news: Empirical studies of German news media* [Doctoral dissertation]. Utrecht University. https://doi.org/10.33540/1756

Peris, A., Faber, W. J., Meijers, E., & Van Ham, M. (2020). One century of information diffusion in the Netherlands derived from a massive digital archive of historical newspapers: The DIGGER dataset. *Cybergeo: European Journal of Geography*, Article 928. https://doi.org/10.4000/cybergeo.33747

Piotrkowicz, A., Dimitrova, V., & Markert, K. (2017). Automatic extraction of news values from headline text. *Proceedings of the Student Research Workshop at the 15th Conference of the European Chapter of the Association for Computational Linguistics*, 64–74. https://doi.org/10.18653/v1/E17-4007

Potts, A., Bednarek, M., & Caple, H. (2015). How can computer-based methods help researchers to investigate news values in large datasets? a corpus linguistic study of the construction of newsworthiness in the reporting on Hurricane Katrina. *Discourse & Communication*, *9*(2), 149–172. https://doi.org/10.1177/1750481314568548

Specht, D. (2018). Did you find the world or did you make it up? media, communications and geography in the digital age. *Westminster Papers in Communication and Culture*, *13*(2), 1–13. https://doi.org/10.16997/wpcc.298

Swart, J., Peters, C., & Broersma, M. (2017). Navigating cross-media news use: Media repertoires and the value of news in everyday life. *Journalism Studies*, *18*(11), 1343–1362. https://doi.org/10.1080/1461670X.2015.1129285

Törnberg, P. (2024). Best practices for text annotation with large language models. *Sociologica*, *18*(2), 67–85. https://doi.org/10.6092/issn.1971-8853/19461

Trilling, D., Tolochko, P., & Burscher, B. (2017). From newsworthiness to shareworthiness: How to predict news sharing based on article characteristics. *Journalism & Mass Communication Quarterly*, *94*(1), 38–60. https://doi.org/10.1177/1077699016654682

Usher, N. (2020). News cartography and epistemic authority in the era of big data: Journalists as map makers, map users, and map subjects. *New Media & Society*, *22*(2), 247–263. https://doi.org/10.1177/1461444819856909

Vermeer, S., & Groot Kormelink, T. (2024). Netherlands: A diverse media landscape with high audience trust. In A. K. Schapals & C. Pentzold (Eds.), *Media compass: A companion to international media landscapes* (pp. 96–105). Wiley Blackwell. https://doi.org/10.1002/9781394196272.ch10

Vogler, D., Weston, M., & Udris, L. (2023). Investigating news deserts on the content level: Geographical diversity in Swiss news media. *Media and Communication*, *11*(3), 343–354. https://doi.org/10.17645/mac.v11i3.6794

Won, M., Murrieta-Flores, P., & Martins, B. (2018). Ensemble named entity recognition: Evaluating NER tools in the identification of place names in historical corpora. *Frontiers in Digital Humanities*, *5*, Article 2. https://doi.org/10.3389/fdigh.2018.00002

Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2024). Can large language models transform computational social science? *Computational Linguistics*, *50*(1), 237–291. https://doi.org/10.1162/coli_a_00502